WELL WORKOVER PROGRAM

AZADEGAN OIL FIELD

WELL WORKOVER#1 PROGRAM

AZNS-166 (PAD #09), TYPE: IK

Drilling Department

DECEMBER 2024

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1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun province on a large long-axis anticline with N-S trending with approximate dimensions about 20 km width and 60 km length near to Iran-Iraq border, to the east of Yaran Oil Field.

The trap area of the south Azadegan is limited to 43.2x17.2 km. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643m. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340m (Neyriz Fm.), mainly to evaluate the Jurassic formations.

Based on the exploration wells results, carbonate reservoir of Sarvak Formation is broadly developed in Azadegan Field. This formation was penetrated in all of seven exploration wells (AZN-1, AZN-2, AZN-3, AZN-4, AZN-5, AZN-6 and AZN-10). Clastic oil-bearing zones of Kazhdumi Formation developed in main anticline of Azadegan Field, extending from north to center of South Azadegan. The Clastic oil-bearing zones of Gadvan Formation developed only in crest area and south parts of South Azadegan Filed, which was discovered by AZN-1 in crest and AZN-4 in south parts. The carbonates of Fahliyan Formation also developed in Azadegan field, but the oil reservoir of this formation developed in South part of South Azadegan.

The Azadegan Field is an elongated anticline, trending in N-S direction, belonging to "Arabian Trend".

The stratigraphy sequence of the field includes AghaJari, Gachsaran, Fahliyan, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan and Fahliyan Formations, from top to bottom. This sequence in well no AZN-10, the deepest up to now drilled well of the field, continues with Garau, Gotnia, Najmah, Sargalu, Alan, Mus and Neyriz formations.

The Dominant lithofacies of Sarvak Formation are Rudist-bearing packstone and partially grainstone, which contains abundant bioclasts. Some other facies like matrix-dominated wackestone and mudstone are also found.

The AZADEGAN Oilfield is Located in Khuzestan Province and is Approximately Located 80 Kilometers West of Ahwaz City, near to Iran-Iraq Border. It is in The Northwest of The YADAVARAN Field and in The West of JUFEYR Field. The Average Monthly Temperature is 45ºc in summer, and Highest Temperature is 52ºc. The Lowest Temperature is -4.4ºc in winter. Maximum Rainfall is 78 mm in February, Winter Season.

This Document covers the work over of well AZNS-166. Main headlines of program are de-complete the well and run new completion string.



Figure 1: AZNS Oilfield Location Map

1.2 Well Information

Table 1: AZNS-166 information

Country / Area Name:	Iran / Khuzestan
Client	Petro Pars Limited
Field Name:	Azadegan
Type Of Operation:	Workover
Well Name / Type :	AZNS-166 / Kazhdumi well in Block I
Pad Number:	09
Well Type:	Deviated
Wellhead Coordinate:	N: 3,472,595 E: 196,992
Kazhdumi E.P. Coordinate:	N: 3,472,112.2 E: 196,261.9 Z: -3357.4m TVDss
TD Coordinate:	N: 3,471,897.1 E: 195,955.7 Z: -3561.00m TVDss
Well Objective:	De-complete the well, Re-complete the well,
RTE:	12.95 m
GLE:	3.95 m
Well Duration :	44.2 days (Main Plan) + 95 Days (Contingency Plan)
Total Depth (MD):	4280 mMD
Producer Formation:	Kazhdumi (4 - 1/2" Sloted Liner)
Formation TD :	Kazhdumi
Reservoir :	Kazhdumi

1.3 Area Distribution

According to the drilling history, the CONTRACT AREA is distributed into three blocks as per the geological features with regard to DRILLING OPERATION. This distribution may differ according to knowledge accumulated in the future WORK.

Contract area is as per below:

X (m)	Y (m)
185640	3434275
188020	3477361
205224	3476408
202842	3433327

Table 2: Area Distribution

Block I is located on the north part of the CONTRACT AREA in north of well AZN-010. Gachsaran formation is characterized by abnormal high pressure in this block.

Block II is located between well AZN-010 and AZN-011. In this block, Gachsaran formation is characterized by pressure tending to normal

Block III is located on the south of line well AZN-011. In this block, Gachsaran formation is characterized by normal pressure and all drilled wells in this block suffered from complete loss at Pabdeh formation.

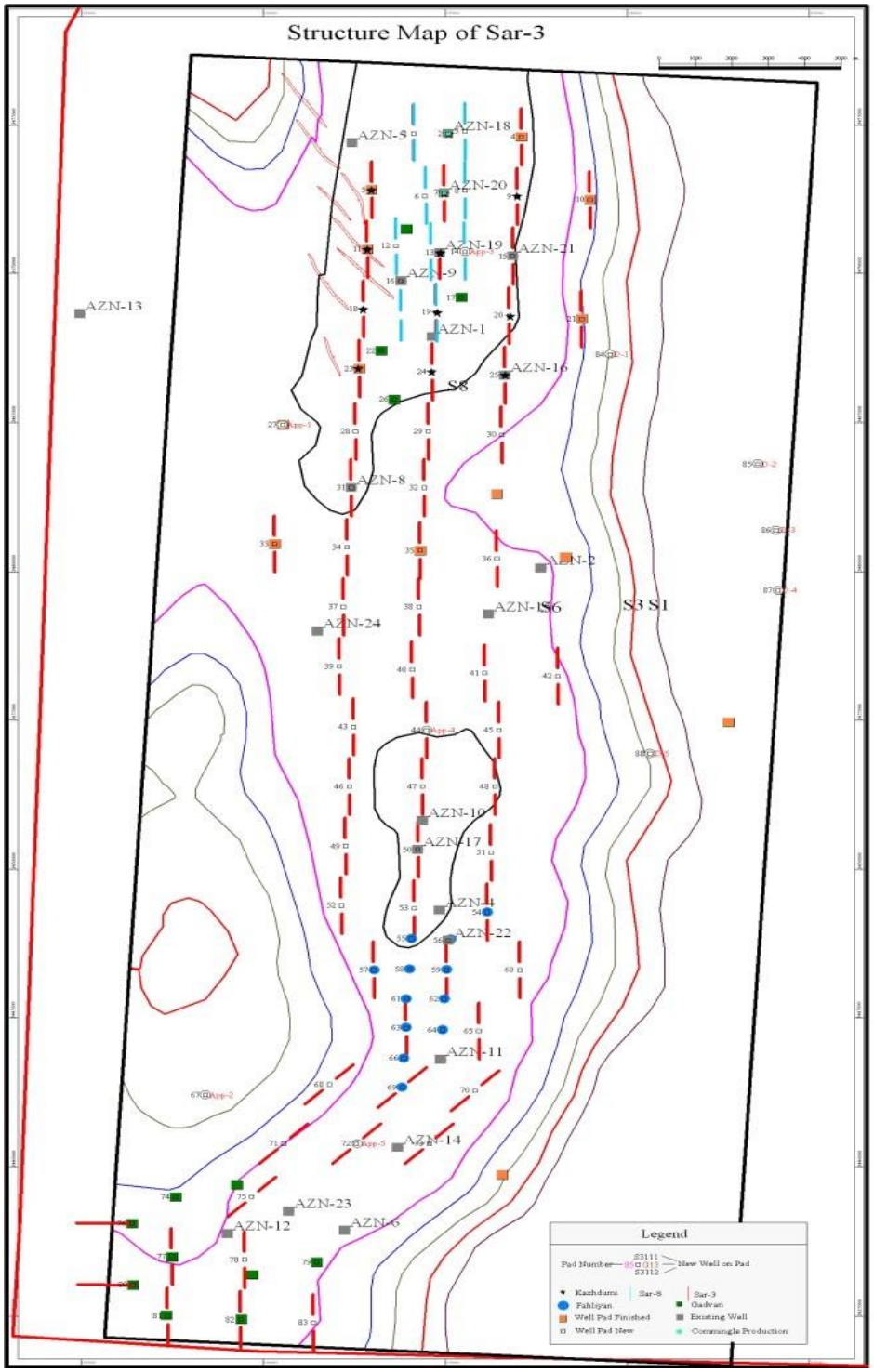


Figure 2: AZNS Area Distribution

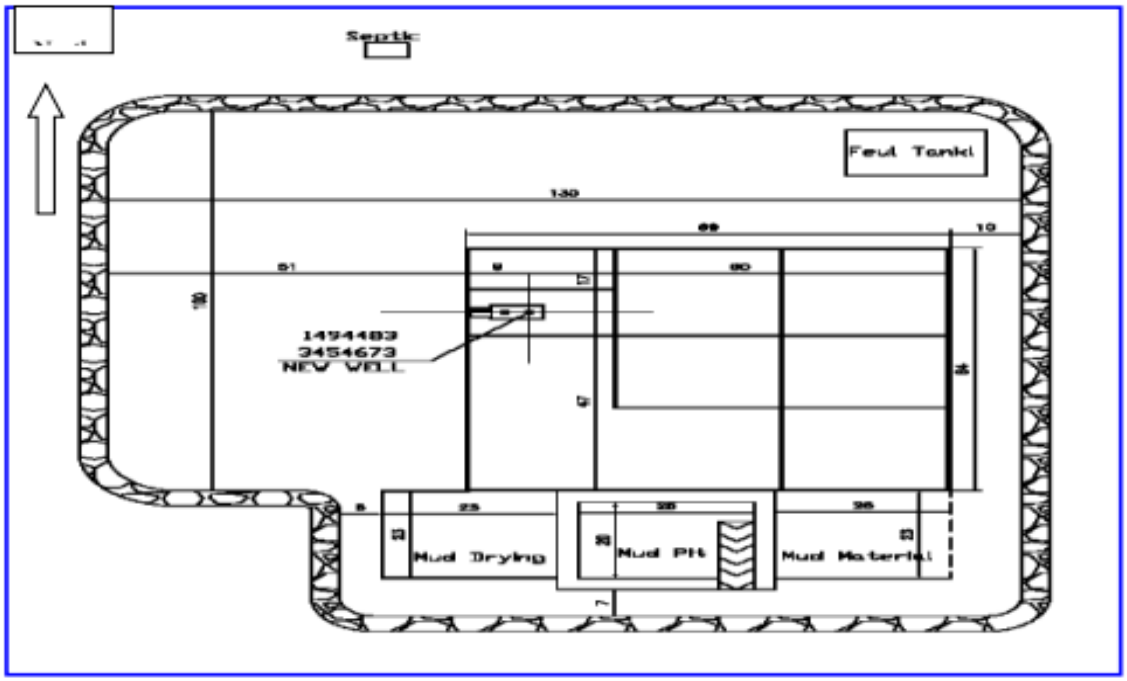
1.4 Well Pad Specification

Well Pad Type	Well Pad Dimension (m)	Mud Pit Dimension (m)	Cellar Dimension (m)	Conductor Size
Single Well	130 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D	30"
Double Well	145 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D	30"
Triple Well	160 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D + 4.0L x 3.8W x 3.4D	30'

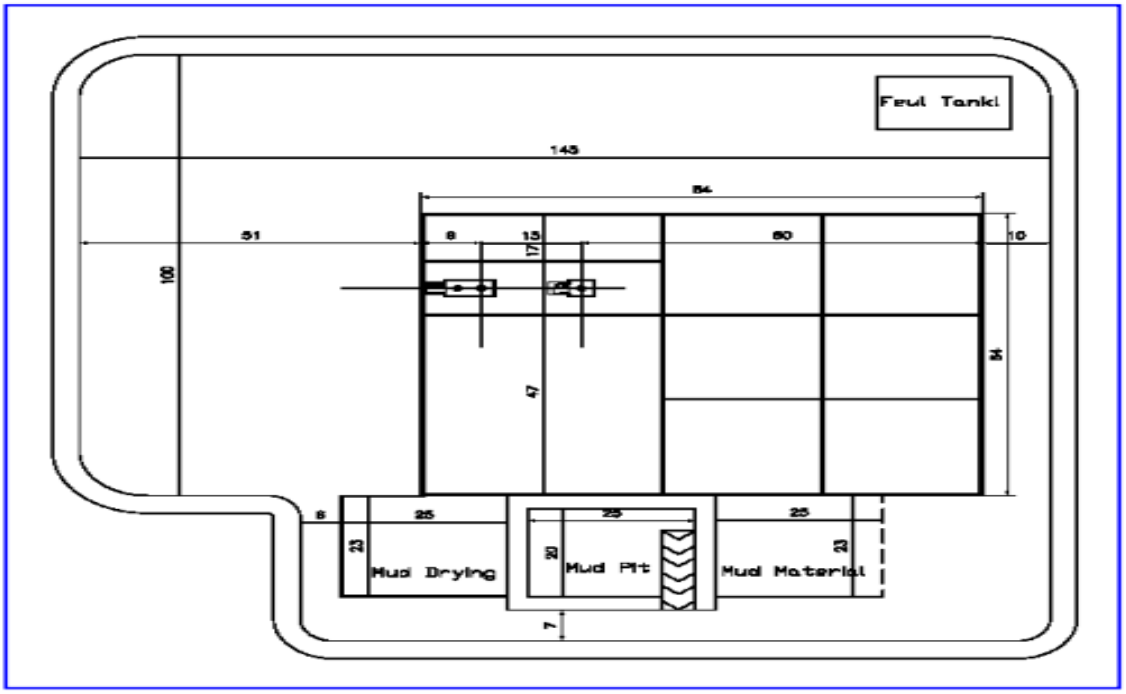
Table 3: Well Pad Specification

Compressive Strength for reinforced concrete after 28 days: 280 kg/cm2
• Fuel tank concrete pond dimension: Length 20m×Width 12m×Height 0.5m
• Drainage: Width 30cm × Depth 30cm
• Burning pit: 15m×15m square, located on South-East or South-West with at least 350m distance from well head
• Cellar to cellar distance in multi-well well pad: 15m
• Conductor pipe depth under cellar bottom is 5.5m. The conductor pipe will be driven while constructing the well pad.
• Mud material dimension is 25m*35m +granular surface
• Mud drying dimension is 25*35m + concrete surface
• Well pad surface is gravel. Access road granular surface. The level of well pads is 3.95 MSL

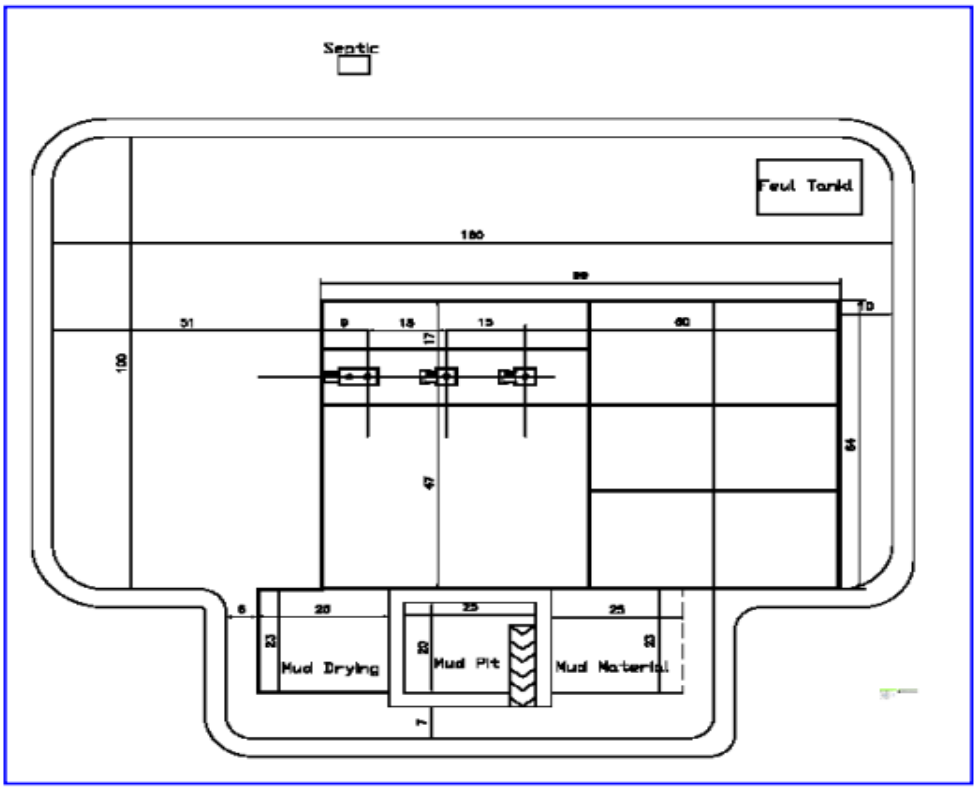
1. Types of well pads:
• One well in one pad.
• Two wells in one pad. 2 Sarvak horizontal wells or 1 Sarvak horizontal well plus 1 vertical/directional well).
• Three wells in one pad: 2 Sarvak horizontal wells and 1 vertical/directional wells.



Well Pad for one well



Double-well Well Pad



Triple-well Well Pad

Figure 3: Well Pads Sketch

1.5 Reservoir Data

Reservoir Name:	Kazhdumi	Unit
Fluids:	Oil	….
Datum Depth:	3480	M,TVDss
Datum Pressure:	KAZ2= 4900 KAZ4= 4900	Psi
Pressure Gradient:	0.31	Psi/ft
Temperature:	235	F
Temperature Gradient:	1.32 0.73 2.4	F/100ft C/100ft C/100m
H2S:	0	PPM
CO2:	41000	PPM
Salt Salinity :	180000	ppm
Calcium Content:	2911	(mg/lit)

Table 4: Reservoir Data

Reservoir	Sub-layer	Mid of Sub-Layer Depth in AZNS-166 (mss)	Temp (°F)	Temperature Gradient (°F/100ft)	Current Reservoir Pressure (psia)	Basis for Reservoir Pressure	WOC (mss)	TTZ (mss)
Kazhdumi	Kaz-Top	-3450	238	1.3	-	-	-3595	-3589
	Kaz-1	-3512	244.7		-	-		
	Kaz-2	-3526	244.7		4080	Dynamic Model		
	Kaz2_2	-3539	244.7		4061	Dynamic Model		
	Kaz-3	-3547	244.8		4175	Dynamic Model		
	Kaz-4	-3550	245.3		4184	Dynamic Model		
	Kaz-5	-3552	246		4030	Dynamic Model		

Table 5: Predicted Pressure and Temperature and TTZ Contact

1.6 Formation Pressure Prediction in Block I

Predicted Formation Pressure in Block I				
Formation	**Interval**	**Pore Pressure**	**Collapse Pressure (Break-Out Pressure)**	**Fracture Pressure (Break-Down Pressure)**
	m	*SG*	*SG*	*SG*
Aghajari	0～1368	0.98～1.15	0.88～1.33	1.01～1.89
Gachsaran	1368～1957	1.25～1.80	1.03～1.82	1.89～2.01
Asmari	1957～2284	1.14～1.26	0.6～1.28	1.43～1.79
Pabdeh	2284～2458	1.12～1.26	0.561～1.36	1.40～1.78
Gurpi	2458～2533	1.12～1.25	0.66～1.02	1.40～1.76
Ilam	2675～2749	1.13～1.18	0.24～1.12	1.41～1.72
Laffan	2749～2756	1.13～1.18	0.24～1.12	1.41～1.72
Sarvak	2756～3372	1.14～1.26	0.24～1.11	1.44～1.81
Kazhdumi	3372～3593	1.1～1.14	0.46～0.96	1.93～1.94
Dariyan	3593～3771	1.18～1.25	0.55～1.10	1.96～2.06
Gadvan	3771～4119	1.1～1.25	0.55～1.44	1.97～2.11
Fahliyan	4119～TD	1.65～1.3	1.20～1.75	2.08～2.12

Table 6: Formation Pressure Prediction

1.7 Well History

1.7.1 Drilling History

The AZNS 166- (Pad 09) was drilled by rig 28 Fath, which rig moving was spudded on 24-Janury-2019 @ 06:00.

- The Well AZNS-166 was spud in with 26” bit at 06:00 on Jan 24, 2019, Drilling of this section was started from Surface on Jan 24, 2019 in Aghajari formation and ended drilling at 149m (C. Point). Install New C.H.H 20-3/4" 3K & Test P-Seals W/ 500 PSI for 15 Min.20" casing was run to 155m & cemented by inner stab-in string with slurry to surface.
- Drilling of 17.5” section was started from 149 MT on Jan 31, 2019 in Aghajari formation to 1338 m (13-3/8" CSG Shoe) and Install 20-3/4"-3K * 13-5/8"-5K C.H.S (MSP DRILEX, S/N: CS1523N140058).
- The 12 ¼’’ hole was started from 1338m on March 5, 2019 with 115 pcf mud weight in Gachsaran formation to 1343. then increase mud weight gradually to 124 pcf to 1765 m. observe 2-4 BPH gain and cut mud weight return to 112 PCF and increase Ca2+ ion to 6000 mg/lit. close the well and prepare 650 bbl with 130 PCF mud and kill the well and then RIH to bottom and balance mud weight at 130 PCF with no loss & no gain. Run 9 5/8” casing to 1764 MT and cementing (397 bbl 138 PCF as lead and 118 bbl 125 PCF as tail). Install 13-5/8"-5K x 11"-10K T.H.S (MSP/DRILEX).
- The 8 1/2’’ hole was started from 1764m on March 26, 2019 in Gachsaran formation. With 78 PCF mud weight to 1765.5m. Continue drilling Resume Drlg Formation to 2273m. Start Dir.Drlg from kop 2735 to 4197m (7" Liner Point) While Displace Kcl-Polymer Mud (81 PCF) In Hole W/ Oil Base Mud (80 PCF).
- The 6 1/8’’ hole was started from 4197m on August 31, 2020 in KAZHDUMI Formation with 82 PCF Mud W/ 6-1/8 PDC Bit & Dir. Drilling BHA to 4280m (TD). RIH W/4-1/2” Slotted Liner to 4280m Set HGR. With max inclination 63.5 and AZ 235.
- Displace Mud in Well W/ Comp. Fluid (MW: 82 PCF). RIH 3-1/2" Completion String as Program to 2776mWhilePress.Test Comp. String Several Steps W/ 4500 PSI / 15 MIN, OK. cont. RIH Comp. String to 2785.2m, Land TBG HGR in 11"-10K T.H.S, Press. Test Comp. String W/ 4500 PSI / 15 MIN. Run in Comp. String W/ SB Pulling & Plug to 2770.1m Set Prong in NO-GO. Pressure test Comp. String W/4500psi for 15MIN, Ok. Close 3SV, laydown41/2"L. JT, install 4"TWCV in TBG HGR, P/T TWCV & TBG HGR W/ 4500 Psi, OK, N/D BOP Stack, Install 11"*4 1/16"-10K BTM Master Valve.

  RIH W/ GS Pulling & Remove NO-GO. RIH C.T.B to 400 M, Obs. Increase Well Head Press. To 200PSI, Close The Well, Monitored Well After 3 Hr, Press: 460 PSI, Open The Well, While Flow the Well, Cont. RIH Coiled Tubing to 4280m Displace Comp. Fluid in Well W/ 240 BBL Diesel. Cont. Flow The Well W/ Chock Sizes: 54 & 48 & 32/64", Oil

Weight: 54.5 LB/FT3, BS&W: 0, Ann. Press: 750 PSI, In Order to Monitor the Well & Ann. Press., Close The WELL @ 24:00 Monitored The Well & Ann. Press., Well Head Press.: 1400 PSI, ANN. PRESS: 0 PSI.

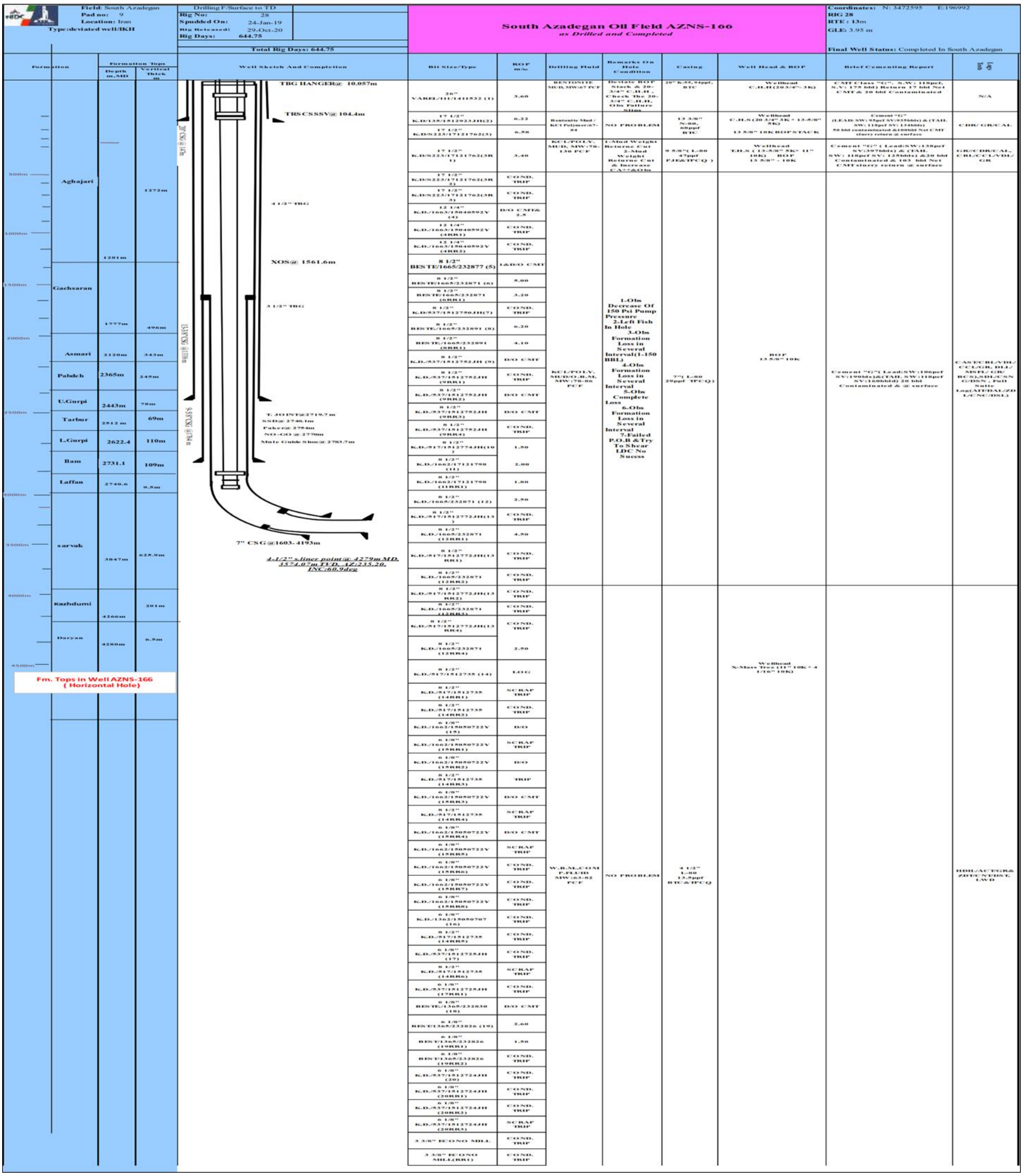
Again RIH by Slick Line to NO-GO, Install Plug & Prong in NO-GO Landing Nipple, Test W/ 2500 PSI, AFTER 3 TIMES, Finally Ok, Again RIH by Slick Line & Install Protection Sleeve Inside The 3SV. IH Coiled Tubing to 2770m Displace Oil in Well W/ Diesel, R/U H-Lubricator Install 2-WAY Inside the Tubing HGR, L/D H-Lubricator, Close Top, Kill, Production & Chock Valve of X-Mass Tree. Release The Rig @ 24:00 on October 29, 2020 (1399/08/08).

1.7.2 Operation after Completion History

- Well has secured with installing TWCV, plug & prong (in No-Go Landing Nipple) & closed 3SV (CL pressure: 0psi)
- Retrieve two way BPV, rubber packing has been damaged.
- RIH to retrieve prong unsuccessful, RIH with bailer gauge to 2760m it was full of Asphaltene (2 times).
- Pressurized CL with 4500psi & opened 3SV, RIH CT to depth of obstacle without pumping fluid, CT couldn't pass SSD depth @ 2740.1m.
- RIH CT to 2769.9m (No-Go Depth), pumped 15bbl diesel & spotted 15bbl Xylene on obstacle, retrieve prong coated with Asphaltene.
- RIH CT with GS 2.5" without pumping fluid to No-Go depth @ 2769.9m & apply 17klb over-pull to retrieve plug. CT couldn't pass 3SV, applying 4klb over-pull, plug was retrieved from fishing neck. Moreover, impression block of 3SV flapper observed on neck of plug.
- SIWHP = 1700psi, flow the well (FWHP = 700psi), close 3SV flowing continued, RIH SL with LIB while 3SV flapper is closed, mark of 3SV flapper observed on LIB, flow the well towards burn pit while 3SV flapper was closed, well produce non-stop, close production valve, open 3SV flapper & RIH SL with LIB @ 102.5m, dipping valve & jar down mark on 3SV flapper observed on LIB, RIH SL with LIB while 3SV flapper was closed. Mark of 3SV flapper observed on LIB, RIH SL with 1-1/2" stem while 3SV flapper was closed. Weight dropped @ 102.5m, open 3SV flapper, RIH against SL to 102.5m & observed weight dropped.
- RIH with 2.8" LIB several times while 3SV closed or open & observed mark on it but in close position pressure dropped & remained constant. (600 – 800psi)
- WHP = 1500psi, RIH SL with special protection sleeve (profile: AF, ID: 2.7, OD: 3.437", modified by PED.CO). SL @ 324ft. and try to pass 3SV flapper tip of the P.S pass the flapper but not completely, effects were observed on the tip of P.S.

- RIH 2.5” bailer, 2.3” LIB, 1-7/8” taper tap, couldn’t pass from 330ft., RIH with 2.320” LIB to 9090ft. (observed some drag during RIH probably due to presence of Asphaltene), observed plug impression on the LIB.
- RIH SL with tubing swage 2.56”, SL @ 330ft. TOF, observed drag, jar down to pass this depth & again jar up, repeated that for several times, no success.
- RIH SL with taper tap 1.5”, SL @ 9090ft. TOF & tried to catch the plug, R-pulling pin was sheared with 1200lb at the depth of 9090ft. (rope socket 1-7/8” + 3ft. stem 1-7/8” + taper tap 1.5”) were left in the well as fish. RIH SL with R-pulling 1-7/8” to 9090ft. TOF & tried to catch the fish, flow the well meanwhile trying to catch the fish, retrieved successfully.
- Repeated RIH SL with taper tap 1.5”, SL @ 9090ft. TOF & tried to catch plug, R-pulling pin was sheared with 1200lb at the depth of 9090ft. (rope socket 1-7/8” + 3ft. stem 1-7/8” + taper tap 1.5”) were left in the well as fish. RIH SL with R-pulling 1-7/8” to 9090ft. TOF & tried to catch the fish, flow the well meanwhile trying to catch the fish, retrieved successfully. Secured the well.
- Install WH & XMT Acc.
- At the presence wellhead, accessories disconnected.

1.8 Well Montage before Workover#1



Bit Size/Type	ROP m/hr	Drilling Fluid	Remarks On Hole Condition	Casing	Well Head & BOP	Brief Cementing Report	
26" VAREL/111/1411532 (1)	3.60	BENTONITE MUD, MW:67 PCF	Deviate BOP Stack & 20-3/4" C.H.H ; Check The 20-3/4" C.H.H, Obs Failure Situa	20" K-55, 94ppf, BTC	Wellhead C.H.H (20 3/4"- 3K)	CMT Class "G", S.W: 118pcf, S.V: 175 bbl) Return 17 bbl Not CMT & 20 bbl Contaminated	N/A
17 1/2" K.D/135/1512932JH(2)	6.22	Bentonite Mud / KCl Polymer:67-84	NO PROBLEM	13 3/8" N-80, 68ppf BTC	Wellhead C.H.S (20 3/4" 3K * 13-5/8" 5K) 13 5/8" 10K BOP STACK	Cement "G" (LEAD SW: 95pcf SV:935bbls) & (TAIL SW: 118pcf SV: 134bbls) 50 bbl contaminated &100bbl Not CMT slurry return @ surface	CDR/ GR/CAL
17 1/2" K.D/S223/17121762(3)	6.58						
17 1/2" K.D/S223/17121762(3R 1)	3.40	KCL/POLY, MUD, MW:178-130 PCF	1-Mud Weight Returns Cut 2-Mud Weight Returns Cut & Increase CA++&Obs	9 5/8"(L-80 47ppf PJE&TFCQ)	Wellhead "EHS (13-5/8" 5K* 11" 10K) BOP 13 5/8" - 10K	Cement "G" (Lead:SW:138pcf SV:397bbls) & (TAIL SW: 118pcf SV: 128bbls) &20 bbl Contaminated & 165 bbl Not CMT slurry return @ surface	GR/CDR/CAL, CBL/CCL/VDL/ GR
17 1/2" K.D/S223/17121762(3R 2)	COND. TRIP						
17 1/2" K.D/S223/17121762(3R 3)	COND. TRIP						
12 1/4" K.D./1663/150405925Y (4)	D/O CMT& 2.8						
12 1/4" K.D./1663/150405925Y (4RR1)	COND. TRIP						
12 1/4" K.D./1663/150405925Y (4RR2)	COND. TRIP						
8 1/2" BESTE/1665/232877 (5)	1&D/O CMT						
8 1/2" BESTE/1665/232871 (6)	5.00						
8 1/2" BESTE/1665/232871 (6RR1)	3.20						
8 1/2" K.D/537/1512750JH(7)	COND. TRIP		1-Obs Decrease Of 150 Psi Pump Pressure 2-Left Fish In Hole 3-Obs Formation Loss In Several Interval(1-150 BBL) 4-Obs Formation Loss In Several Interval 5-Obs Complete Loss 6-Obs Formation Loss In Several Interval 7-Failed P.O.B & Try To Shear LDC No Sucess				
8 1/2" BESTE/1665/232891 (8)	6.20						
8 1/2" BESTE/1665/232891 (8RR1)	4.10						
8 1/2" K.D./537/1512782JH (9)	D/O CMT						
8 1/2" K.D./537/1512782JH (9RR1)	COND. TRIP	KCL/POLY, MUD/O.B.M, MW:78-86 PCF		7"(L-80 29ppf TFCQ)	BOP 13 5/8" 10K	Cement "G"(Lead:SW:110bpcf SV:190bbls)&(TAIL SW:118pcf SV:160bbl) 20 bbl Contaminated & @ surface	CAS/CBL/VDL/ CCL/GR, DLL/ MSFL/ GR/ BCS),SDL/CSN G/DSN ,/ PulI Suite Log(AITD&AL/ZD L/CNC/DSL)
8 1/2" K.D./537/1512782JH (9RR2)	D/O CMT						
8 1/2" K.D./537/1512782JH (9RR3)	COND. TRIP						
8 1/2" K.D./537/1512782JH (9RR4)	1.50						
8 1/2" K.D./1662/17121798 (11)	2.00						
8 1/2" K.D./1662/17121798 (11RR1)	1.80						
8 1/2" K.D./1665/232871 (12)	2.50						
8 1/2" K.D./517/1512772JH(13)	COND. TRIP						
8 1/2" K.D./1665/232871 (12RR1)	4.50						
8 1/2" K.D./517/1512772JH(13 RR1)	COND. TRIP						
8 1/2" K.D./1665/232871 (12RR2)	COND. TRIP						
8 1/2" K.D./517/1512772JH(13 RR2)	COND. TRIP						
8 1/2" K.D./1665/232871 (12RR3)	COND. TRIP						
8 1/2" K.D./517/1512772JH(13 RR3)	COND. TRIP						
8 1/2" K.D./1665/232871 (12RR4)	2.50						
8 1/2" K.D./517/1512738 (14)	LOG						
8 1/2" K.D./517/1512738 (14RR1)	SCRAP TRIP						
8 1/2" K.D./517/1512738 (14RR2)	COND. TRIP						
6 1/8" K.D./1662/150850722Y (15)	D/O						
6 1/8" K.D./1662/150850722Y (15RR1)	SCRAP TRIP						
6 1/8" K.D./1662/150850722Y (15RR2)	D/O						
8 1/2" K.D./517/1512738 (14RR3)	TRIP						
6 1/8" K.D./1662/150850722Y (15RR3)	D/O CMT						
8 1/2" K.D./517/1512738 (14RR4)	SCRAP TRIP						
6 1/8" K.D./1662/150850722Y (15RR4)	D/O CMT						
6 1/8" K.D./1662/150850722Y (15RR5)	SCRAP TRIP						
6 1/8" K.D./1662/150850722Y (15RR6)	COND. TRIP	W.B.M,COMP.FLUID MW:63-82 PCF	NO PROBLEM	4 1/2" L-80 13.5ppf BTC&TFCQ			BDHL/ACT/GR& ZDTCN/DSI, LWD
6 1/8" K.D./1662/150850722Y (15RR7)	COND. TRIP						
6 1/8" K.D./1662/150850722Y (15RR8)	COND. TRIP						
6 1/8" K.D./1362/1505850707 (16)	COND. TRIP						
8 1/2" K.D./517/1512738 (14RR5)	COND. TRIP						
6 1/8" K.D./537/1512725JH (17)	COND. TRIP						
8 1/2" K.D./517/1512738 (14RR6)	SCRAP TRIP						
6 1/8" K.D./537/1512725JH (17RR1)	COND. TRIP						
6 1/8" BENTE/1365/232830 (18)	D/O CMT						
6 1/8" BENT/1365/232826 (19)	2.60						
6 1/8" BENT/1365/232826 (19RR1)	1.50						
6 1/8" BENT/1365/232826 (19RR2)	COND. TRIP						
6 1/8" K.D./537/1512724JH (20)	COND. TRIP						
6 1/8" K.D./537/1512724JH (20RR1)	COND. TRIP						
6 1/8" K.D./537/1512724JH (20RR2)	COND. TRIP						
6 1/8" K.D./537/1512724JH (20RR3)	SCRAP TRIP						
3 3/8" ECONO MILL	COND. TRIP						
3 3/8" ECONO MILL(RR1)	COND. TRIP						

PEDEC

Drilling History

26" Hole Section: The Well AZNS-166 was spud in with 26" bit at 06:00 on Jan 24, 2019 and ended drilling at 149m (at 03:00) on Jan 26, 2019.. This hole was started and finished by using Bentonite mud with 66-67 PCF from surface to TD section, with NL-NG.. 20" casing (K-55, 94#, BTC) was run to 147m and cemented. Install C.H.H. 20-3/4"- 3K (MSP Drilex, BTM slips types).Nipple up20-3/4"- 3K BOP STACK (20-3/4"-3K SPACER SPOOL + 20 3/4"- 3K BLIND RAMS+ 20 3/4" - 3K PIPE RAM + 20 3/4" -3K HYDRILL) and tested it. RIH w/ 17-1/2" Rock bit to 146m, tag CMT & D/O CMT & F.Shoe & Rat hole f/146 to 149m.

17 1/2" Hole Section: Drilling of this section was started from 149 MT in Aghajari formation and continued to 346 MT with bentonite type mud without loss in formation, while control mud properties with run solid control equipment's. Prepare 850 bbl new KCL Polymer mud (mud weight: ٧٥ PCF and Vis.:39 Sec/qr.), and displace with Bentonite mud. Drilling operation resumed to 955 MT with 0-2 BPH formation loss, while increase mud weight from 75 PCF to 77 PCF as program, prepare Hi Vis. Pill with XC Polymer and circulate for good hole cleaning. Stop operation as force major due to level up of water around location. Resumed drilling to 1338 MT (TD section) on 1397/12/02 while increase mud density to 80 PCF several stages as per program with 0-2 BPH formation loss. Check hole condition, increase mud density to 84 PCF several stages due to observe tight hole with no loss & no gain and work on mud properties according to the program. RIH w/ logging tools CDR/GR/CAL to 1336.5m, Log Up to 140m (20" CSG Shoe @ 147m). POOH to surf. 13 ⅜" casing (68#, N-80, BTC) was run to 1338m (13-3/8" CSG Shoe) and cement job was performed, obs bumping w/ 1500 psi, hold 15min, OK. Check back flow, no return. While get return CMT during dumping (20 bbl Mud Contaminate + 100 bbl Preflush + 50 bbl CMT Contaminate+ 100 bbl Net CMT + 40 bbl loss). Install 20-3/4"-3K * 13-5/8"-5K C.H.S (MSP DRILEX, S/N: CS1523N140058). Nipple up 13 5/8" 10K BOP and manifold and tested them. RIH w/ 12-1/4" PDC bit to 1309m, string took weight, try to wash down, no succ. D/O CMT plugs. Cont. D/O CMT Plugs med. to hard CMT & F.Collar f/1309 to 1314m, Cont. D/O shoe track CMT to 1331m (7m above 13-3/8" CSG shoe) . Displace KCL polymer mud 84pcf in hole w/ HMW 115pcf, perform circulation. Cont. D/O shoe track CMT 13-3/8" CSG shoe @ 1338m & rat hole.

12 1/4" Dirc. Hole Section: The 12 ¼" hole was started from 2215m on June 07, 2018 to 2240m. RIH w/ CBL/VDL/CCL/GR logging tools to 2215m (13-3/8" CSG shoe @2214m), Log up to 100m (57m above 20" shoe), Resume drilling f/2240 and ended at 2790m on July 06, 2018 (KOP@ 2295m). In this phase following formations were drilled: Tarbur Member (Top @ 2326M), Lower Gurpi Member (Top @ 2406M), Ilam (Top @ 2555M), Laffan (Top @ 2697M), Sarvak Zone – 1 (Top @ 2709M), Sarvak Zone – 2 (Top @ 2756M), and Sarvak Zone – 3 (Top @ 2783M). KCL polymer mud was used in the phase. Average ROP was 30 min/m, average WOB was 19klb and average SPP was 1980psi. 9 5/8" casing (L-80, 47#, TPCQ) was run to 2790m and cement job was performed. After pumping 645 bbl mud obs bumping w/3000 psi, hold 15 min , OK as CIT test, release press check back flow no return. Nipple up BOP and manifold and tested them according to relative criterion. Install 13-5/8"-5K * 11"-5K T.H.S. (MSP DRILEX), cont. N/U 13-5/8" BOP. RIH w/8-1/2" B.Bit to 2767m. D/O Top & Btm plug, F.collar & medium to hard CMT to 2787m (3m above 9-5/8" CSG shoe), displace mud in hole f/87 to 75pcf KCL-Polymer (step by step). Cont. D/O cmt & 9-5/8" CSG shoe @2790m & rat hole to 2790.6m.

8 1/2" Hole Section: The 8 1/2" hole was started from 1764m on March 26, 2019 in Gachsaran formation with making with 78 PCF mud weight to 1765.5m. Stop Drlg, Force Major Due To Come Up The Water Level OF Lake (33 HRs) , Start POOH F/ 1765.5 TO 1706m .Resume Drlg F/ 1765.5 TO 1767 M Perform FIT w/Rate 0.2 BPM, Max Press: 320psi, Static for 15 min, CUM. VOL.: 0.5 bbl. Release pressure, Cum Back flow : 0.5bbl, (EMW:86pcf). 'POOH F/ 1767 TO surface. RIH w/ CBL/VDL/CCL/GR logging tools to 1767m, Log up to 1238m (100M ABOVE 13-3/8"CSG SHOE) POOH TO SURF. RIH to 1750 M While Slow Circulation. Wait on PEDEC Due To Don't Have Any Program. RIH to 1767m Resume drilling to/1930m OBS. Decrase Of 150 PSI Pump Pressure, Stop Drlg ,POOH TO 150m While OBS Wash Out @ 220m (Between Connection Of Last HWDP & First 6-1/2" D.C). RIH to 1930m Resume drilling to/2109m W/ 3-4 BPHL . Suddenly Drop S.p.p Pressure And String Weigh Cut.POOH to Surface. Obs Twist OF Strg.RIH W/ 7 7/8" OVERSHOT 1960.5 M (TOP OF FISH), Retrieve Fish , POOH to Surface Retrieve A part of fish (TOTALL LENGTH OF FISH: 9.7 M, TOP). RIH W/8-1/2" B.BIT 2096 While W&R Several point and AT 2096 M, Precaution Push Fish TO 2099.6 M. Weight Up Mud F/ 78 TO 80. POOH To Surface. RIH W/ 7 7/8" OVERSHOT 2099.6 M (TOP OF FISH), work on fish POOH To Surface Retrieve 100 % Fish. RIH W/8-1/2" PDC.BIT To 2109, Resume Drlg Formation To 2213m. 'Stop Operaion Announce Force Majeure Due To Increase Water Level Around Location Probability Incoming To Locaion, POOH To 1720m(54 M ABOVE 9 5/8" CSG SHOE) Close The Well.(From 3 O`clock on April 10, 2019 to July 5,2019). RIH TO 1810mW/W&R. POOH To Surface. RIH W/8-1/2" PDC.BIT To 2213, Resume Drlg Formation To 22735m.While The Avarage 3-8 BPH & 10-30 Gain. And displace Mud In Hole In Two St-eps (F/82 To 83 PCF & 83 To 84 PCF), POOH To Surface. RIH W/ 6-1/8" C.CUT BIT To 2320m OBS 60 BPH Loss. RIH To 2375m Perform Cementing Operaion (50 BBL CMT Slurry WT: 93 PCF).POOH To 1764m (9-5/8" CSG SHOE) WOC, RIH To 2344m Tag CMT, Try To RIH With Wash, NO Succ. RIH To 2334m Perform Cementing Operaion (50 BBL CMT Slurry WT: 93 PCF) POOH To 1764m (9-5/8" CSG SHOE) WOC, RIH To 2236m Took Weigh Wash Down To 2238m And AFter That Try To Wsh Down NO Succ. POOH To Surface. RIH W/8-1/2 B.BIT To 2238m Took Weight Star D/O CMT TO 2376m ,RIH O TO 2729M, Took Weight, W&R 2735m. C.B.U & Detect Max 100% TG ON BTM'S Up, MW Cut To MIN75PCF, CIRC. & Balance MW To 84 PCF. POOH To Surface. RIH W/ Logging Tools ('RUN#1 RESISTIVITY & SONIC (DLL/ MSFL/ GR/ BCS) To 2687m While Log Up To 1764m POOH To Surface , RIH W/ Logging Tools (' RUN#2 CDR/GR) To 2660m While Log Up To 1764m POOH To Surface. RIH W/8-1/2" B.BIT TO 2729m Took Weight, W&R To 2735m. C.B.U & Detect Max 14% TG On BTM'S UP, MW Cut To Min 81PCF, CIRC. & Balance MW To 84PCF W/ 8-10 BPH LOSS. POOH To Surface. RIH W/ Logging (RUN#3 SDL/CSNG/DSN) Tools To 2670m While Log Up To 1764m. POOH To Surface. RIH W/ 6-1/8" C.CUT BIT To 2360m OBS 20 BPH Loss.Perform Cementing Operation (50 BBL CMT Slurry WT: 93 PCF) , POOH To 1880m,Close The Well WOC. OPEN The Well Pump 140 BBL MUD (84 PCF), OBS NO RETURN ,POOH To 1764 MT PUMP 50 BBL MUD (84 PCF) Got Return In Annulus, OBS Mud Level Drop Rapidly, Close The Well, RIH To 2285m Took Weight POOH To 2270m Pump 10 BBL CACL2 (84 PCF) + 10 BBL FRESH WATER + 25 BBL SODIUM SILICATE + 10 BBL FRESH WATER + 10 BBL CACL2 (84 PCF) + 10 BBL FRESH WATER, DISPLACE WITH 130 BBL 84 PCF MUD, Close The Well ,POOH TO 1764 MT .RIH To 2285 m , OBS 80 BPH LOSS POOH To 1764 m. RIH To 2225, Pump 10 BBL CACL2 (84 PCF) + 10 BBL FRESH WATER + 25 BBL SODIUM SILICATE+ 10 BBL FRESH WATER + 10 BBL CACL2 (84 PCF) + 10 BBL FRESH WATER, MIX & PUMP 50 BBL CMT SLURRY (WT: 93 PCF), CIOSE The Well, Open The Well POOH To 1900m Flush String POOH To 1764m Try To Circ. No Succ. POOH To 90m OBS 5 JT 3 1/2" DPs, CONT. POOH To Surface. RIH W/8 1/2" B.BIT To 2285m, D/O Soft CMT To 2350 , Cont. RIH To 2500m, While Increase Mud Loss To 50-70 BPH. RIH To 2550m W/30-50 BPH Loss. POOH To 2300m OBS Complete Loss While Cut Mud Weight To 81 PCF, POOH To 1764m Close The Well. Open The Well RIH To 2360m Spot 80 BBL LCM Pill, POOH To Surface. RIH W/6-1/8" C.CUT To 2320m , Pump 50 BBL BENTONITE-CMT-DIESEL Slurry (81 PCF), RIH To 2360m, W/Out Tag CMT. POOH To 2021m, Pump 32 BBL CMT Slurry (93 PCF, CMT G DELIJAN), Close The Well, Pump 63 BBL CMT Slurry (93 PCF, CMT G DELIJAN), Displace W/ 121 BBL Mud (84 PCF). Open The Well POOH To 1600m. RIH To 2217m , String Took Weight Con. RIH W/ Washing To 2222m Try TO Wash CMT No SUCC. POOH To Surface. RIH To 2222m , Start D/O CMT To 2380m Cont RIH To 2723m, C.B.U POOH To Surface. RIH To 2735m While W&R Several Points. POOH To Surface.RIH W/ 6-3/4" D.H.M. & 8 1/2" B.BIT To 2735m, Start Dir.Drlg To 4197m (7" Liner Point) While Displace Kcl-Polymer Mud (81 PCF) In Hole W/ Oil Base Mud (80 PCF), POOH To Surface. RIH W/ 6-3/4" D.H.M. & 8 1/2" B.BIT To 4197m While Get Log Down By LWD Tools, POOH To Surface. RIH W/ Full Suite Log Tools (AIT/DAL/ZDL/CNC/DSL) To 3460m While Get Log Down F/ 2575 To 3460 , POOH To Surface. RIH As Scrap Trip To 1730m (Scrap F/1600 To 1730M) POOH To Surface. Run 7" Liner (TOTALLY 226 JTs) to 4193.5m Try To RIH W/ Wash & Rotate LNR No Success. Tag & Confirm Btm (3.5 m Defference W/ Drilled Depth) , L/U CMT Line Set Liner Hanger (7" LNR LAP: 1608.2 M, 7" LNR SHOE: 4193 M) Try To Shear LDC No Succss . 'Weight Of String, Shear s.Tools & Release Setting Tools. POOH To Surface L/D Setting Tools. RIH W/ 6 1/8" PDC BIT To 1613m D/O P.O.Bushing. POOH To Surface. RIH As Scrap Trip To 1700m (Scrap F/1615 To 1715M) , POOH To Surface. RIH W/ 7" RTTS Tools And Try To Set packer & Shear LDC Several Points No Succss. POOH To Surface. RIH W/ 7" RTTS Tools To 1620m , Set Packer @ 1619m Try To Shear LDC W/4500 Psi No Succss. POOH To Surface RIH W/6-1/8" PDC Bit + 4 3/4 " Junk Sub To 4155.4m Took Weight DRLG To 4157.40 (LDC), RIH To 4169m Work On Junk Sub Several Times. POOH To Surface. RIH W/7" RTTS Tools To 1623m Set RTTS PKR. Get Return After Apply 2950 PSI Press OBS. Cut Mud (67 'PCF), Close Top Rams, Monitor SICP, Zero, CIRC.To Get 81 PCF In Return, Open The Well & Cont. Circ. Obs.76 BPH Loss & Get 84 PCF Mud In Return, Circ. Mud To Balanced 80 PCF W/ 25 BPHL, Unset 7" RTTS, POOH 1600m, CIRC. Mud Balance (80 PCF), RIH1623m, Set 7" RTTS @ 1622 m, Circ. Mud And Obs.10 BPH Gain, Cont. Circ. Mud To Balance Mud (80 PCF), F. Check Obs. 4 To1 BPH Gain In Several Times, Pump 40 BBL Gasoil To Check RTTS Packer , F. Check, Obs.3 BPH Gain, Circ. Mud To Balance (81 PCF), 'F.check, Obs.5 To 0.9 BPH Gain In Several Times, Disconnect TDS, Close Top Rams, Pump 40 BBL Gasoil Get Return Gasoil After Pumping 960 BBL Mud, Disconnect TDS,OBS. Flowing Inside The String 0.9 BPH Then Decrease To 0.6 BPH, Close Top Rams, Pump W/ GPM: 50, OBS. 0.5 BPH Flowing Inside The String, Pump W/ GPM: 85, OBS. 0.4 BPH Flowing Inside The String, Connect TDS, Circ. Mud Inside The String, Pump 50 BBL Gasoil Inside The String, Disconnect TDS, After 3 HR OBS 1.5 BPH Return 0.15 BPH Gain From String. Displace Gasoil. Perform Cement Job , Pump 580 SXS Micro SILICA CMT "G" DELIJAN (SW: 106 PCF, SV: 190 BBL) As Lead, Then Pump SILICA FLOUR CMT CLASS "G" DELIJAN (SW: 118 PCF, SV: 160 BBL) As Tail,OBS. 30 BBL Loss During Pumping CMT Slurry & 50 BBL During Displacement, Unset RTTS PKR, POOH To Surface, RIHW/ 8-1/2" B.BIT To 1608m(7" LINER LAP).POOH To Surface. RIH W/ 6-1/8" PDC To 4157m Start D/O CMT To 4160.5m POOH To Surface. RIH W/8-1/2" USED B.BIT + 9-5/8" Scraper To 1608m (Scrape F/1450 TO 1608m) POOH To 1508m Perform7" L.LAP Positive Test W/ 2000 PSI (STEP BY STEP), For 15 Min.OK. RIH To 1608m & Apply 25 KLB Weight 'On 7" Liner Lap To Set Top PKR Of 7" Liner HGR , POOH To Surface. RIH W/ 9 5/8" Dry Test Tools To 1593m While Fill Up 478m String W/ 62.4 PCF Fresh Water As Cousion Set Dry Test Packer @ 1589m, 'Open HYDRO SPRING, Start Initial Flow, OBS. Ann.: Static FOR 3 HRs , While OBS. Bubble For First Of One HR, Close D.C.I.P Valve For 6 HRs Meanwhile Monitoring The ANN. Static. Drop Bar Displace Water Cousion W/52 BBL Mud Unset Packer POOH To Surface. RIH W/6-1/8" PDC BIT To 4600m Tag CMT, D/O Hard CMT & F. Collar TO 4189m POOH To Surface. RIH W/6-1/8" PDC BIT +7" Scrapper To 4189m ('Scrape F/1608M TO 4189m). POOH To Surface. RIH W/ RIH Logging Tools (CAST, CBL, VDL, CCL, GR) To 4183m Log Up To 1608m, POOH To Surface. RIHW/ Logging Tools#1 (CBL ,VDL, CCL, GR) To 4150 Log Up To 1608m, POOH To Surface. RIH W/ Backup Logging Tools#2 (CBL ,VDL, CCL, GR) To 3000 Log Up To 1680m, POOH To Surface. RIH W/ Logging Tools#2 (CBL ,VDL, CCL, GR) To 4183 Log Up To 3800m, POOH To Surface. RIHW/ Logging Tools#1 (CBL ,VDL, CCL, GR) To 4150 Log Up To 3800m, POOH To Surface. RIHW/ Logging Tools (CAST, CBL, VDL, CCL, GR) To 4183m Log Up To 2600m , POOH o Surface. Waiting For Client (PEDEC) Due To Failure Of Road While Slow Cir. W/ 6-1/8 PDC Bit Inside String. POOH To Surface. RIHW/ Logging Tools (CAST, CBL, VDL, CCL, GR) To 4183m Log Up To 14183 Log Up To1608m. POOH To Surface. RIH W/8-1/2 B.Bit + 9-5/8 Scraper To 60m Work on it Several Times POOH To Surface.RIH W/ 9-5/8 RTTS To 400m Set Packer @40m, 'Back Of 5" Dp Top Of RTTS.L/D 13-5/8"-10K Bop Stack Change T.H.S, Install New 13-5/8"-5K x 11"-10K T.H.S Test P-Seal's W/ 3800 PSI, OK, Test Side Valve W/ 7500 PSI,OK. N/U 13-5/8"-10K Bop Stack , Unset 9 5/8" RTTS POOH To Surf. RIH W/6 1/8"B.BIT To 4140 Slow Circ. Inside string. Due To Force Majeure, While Close The Well From @ 20;00 on February 24, 2020(W.H Press, Zero.) To @00;00 on August 24, 2020, Open The Well & Monitor Well Head Press, Zero. POOH To 2343m, OBS. Wash Out Between Connection Of STD No. 74, Cont. POOH To Surface While OBS. Wash Out @ 1650m. RIH W/6-1/8" PDC BIT To 4189m Tag CMT Displace Mud In Well (WATER 63 PCF) W/ Kcl-Polymer (82 PCF), Perform CIT W/3000 Psi OK. Start D/O CMT To 4193& Rate Hole

6 1/8" Hole Section: Perform FIT Test on August 29, 2020, W/ 400 Psi For 15 Min @ 4194m, Ok (PUMP 1 BBL & RETURN 1 BBL),(EMW:87 PCF) .The 6 1/8" hole was started from 4197m on August 31, 2020 in KAZHDUMI Formation With 82 PCF Mud W/ 6-1/8 PDC Bit & Dir.Drlg BHA To 4280m (TD) OBS. PRESS. UP POOH To Surface. RIH W/ Full Suite Log DDL/MSFL/ACT/GR Tools As TLC To 4130m, Latch PWCH In DWCH, Try To Take Signal In Several Times, No Success. POOH To Surface. RIH W/ 6-1/8" PDC BIT+4-3/4" D.H.MOTOR To 4280m As Condition Trip POOH To Surface. RIH W/, DLL/MSFL/DSL/GR Tools As Cable To 4210m, Took Weight, Try To Pass, No Success, POOH To Surface, RIH W/ DLL/MSFL/ACT/GR Tools As TLC To 4208m Try To Pass This Point Several No Success. POOH To Surface. RIH W/ 6-1/8" B.BIT+4-3/4" D.H.MOTOR To 4280m While W&R Several Interval Pump 10 BBL Hi-Vis Pill(KCL-POLYMER MUD) Then Pump 20 BBL Oil Base Mud (82 PCF), Pump 10 BBL Hi-Vis Pill (KCL-POLYMER MUD) Behind 'Displace W/Drlg Mud(WATER BASE MUD, KCL POLYMER, 82 PCF) And Spot O.B.M At BTM POOH To Surface. RIH HDIL/GR Tools As Cable To 4208m, TooK Weight, Try To Pass No Succss. POOH To Surface. 'RIHW/ HDIL/ACT/GR Tools As TLC To 4276m , Log Up To 4193m, POOH To Surface. RIH W/ 6-1/8" B.BIT+4-3/4" D.H.MOTOR To 4280m As Condition Trip , POOH To Surface .RIH W/ ZDT/CNT/DST Logging Tools As TLC To 4277m , Log Up To 4193m, POOH To Surface. RIH W/ 6-1/8" B.BIT+4-3/4" D.H.MOTOR To 4280m As Condition Trip POOH To Surface. RIH W/4-1/2" Slotted Liner To 4280m Set HGR., Shear Ball Seat W/ 2800 PSI, Release Setting Tools, Set PKR, Start POOH To Surface .

COMPLETION

RIH 6-1/8"B.BIT+7" Scraper 2800m(Scrap F/2700 To 2800m) Cont. RIH To 4133m , POOH To Surface.RIH W/ 3-3/8 Econo Mill To 4279m Displace 'Mud In Well W/ Comp. Fluid (MW: 82 PCF) , POOH To Surface. RIH 3-1/2" Completion String As Program To 2776m While Press. Test Comp. String Several Steps W/ 4500 PSI / 15 MIN, OK, Connect 1/4"C/L To TBG HGR, P/T 1/4"C/L W/ 5000 PSI, OK, Bleed Off 1/4"C/L Press To 4000 PSI, Cont. RIH Comp. String To 2785.2m, Land TBG HGR In 11"-10K T.H.S, Press. Test Comp. String W/ 4500 PSI / 15 MIN, OK. R/U Slick Line Equip. Run In Comp. String W/ 2.75"SB Pulling To 2740.1(SSD Depth), After 3 Run Catch Prong, POOH To Surface. Run In Comp. String W/ 2.75"GS Pulling To 2740.1(SSD Depth), Catch 2.75"Plug, POOH To Surface. Run In Comp. String W/ Shifting Tools To 2740.1(SSD Depth), Close SSD After 4 Run. POOH To Surface. L/D Slick Line Equip. POOH To 2783.7m (TBG HGR 1.5 Above T.H.S) Set Ret. Hydraulic Production PKR @ 2754m W/ 2300 PSI, Release Press, Check PKR W/ 20 KLB Overpull ,Ok. R/U Slick Line Equip. Run In Comp. String W/ 2.562"SB Pulling To 2770.1(NO-GO Depth), Catch Prong, POOH To Surface. Run In Comp. String W/ 2.562" Prong To 2770.1(NO-GO Depth), Set Prong In NO-GO POOH To Surface, L/D Slick Line Equip. Press. Test Comp. String W/ 4500 PSI / 15 Min, OK. Land 11"TBG HGR In T.H.S & Half Close Travelling Joint, Tight All TIE-DOWN Screw On T.H.S, Final Press. Test Comp. String W/ 4500 PSI / 15 MIN, OK. Install BOP Of Slickline & Lubricator On Comp. String, Run In Comp. String W/ 2.562"SB Pulling To 2770.1(NO-GO Depth), Catch Prong, POOH To Surface. Run In Comp. String W/ 2.562"GS Pulling To 2770.1(NO-GO Depth), Catch 2.562"Plug, Try To Pull Out Plug In Several Tiems No Succss. Apply 2000 PSI Press In String, OBS. Decrease Press. Run In Comp. String W/ 2.562"GS Pulling To 2770.1(NO-GO Depth), Catch 2.562"Plug, Pull Out Plug To Surface, OBS Left In Hole BTM Part Of 2.562"Plug & Recover Top Of Plug. 'Test PKR & TBG HGR W/ Apply 2000 Psi **Press In** Annulus For 15 Min, OK, Run In Comp. String W/ SB Pulling & Prong To 2770m, Set Prong In NO-GO L.N, POOH To Surf. Apply 300 PSI Press In String, OBS. Decrease Press. Run In Comp. String W/ R-Pulling To 2770m, Catch Prong, POOH To Surf. Run In Comp. String 2 Steps W/ 2.1"Overshot To 2770m Try To Catch Fish No Succss. POOH To Surface. Run In Comp. String Impressin Block To 2770.1m Mark On It POOH To Surface. Run In Comp. String W/ R-Pulling & 2.25 Spear To 2770m, Work On Fish Catch Fish POOH To Surface. 'Test PKR & TBG HGR W/ Apply 2000 Psi Press In Annulus For 15 Min, OK. F/U String W/ 50 BBL Comp. Fluid (CACL2), OBS. No Return, Run In Comp. String W/ 2.562"GS Pulling & 2.532'Plug To 2770.1m Set Plug In NO-GO L.N POOH To Surface. Run In Comp. String W/ SB Pulling & Plug To 2770.1m Set Prong In NO-GO L.N POOH To Surface. F/U String W/40 BBL Comp.Fluid L/D Slick Line Equip. P/T Comp .String W/4500psi For 15MIN, Ok. Close 3SV, L/D 4 1/2"LJT, Install 4"TWCV In TBG HGR, P/T TWCV & TBG HGR W/ 4500 Psi, OK, N/D BOP Stack, Install 11"*4 1/16"-10K BTM Master Valve. Pressure Test C/L W/ 5000 Psi For 15 MIN, OK, 'P/T 2No. S.SEALs On TBG HGR & Ring Gasket & All Tie-Down Screws ON T.H.S W/ 7500 Psi / 15 MIN, OK, Press Test All Vaves Of 4 1/16"-10K X-MASS TREE On Surface W/ 1000-7500 PSI / 5-15 MIN, OK. Install 4 1/16"-10K X-MASS TREE On Bottom Master Valve, Body Press Test X.MASS TREE & METAL SEAL Of TBG HGR W/ 1000-4500 PSI / 5-15 MIN, OK, P/T P-Seals Of 13 5/8"-5K * 11"-10K T.H.S W/ 3500 PSI / 15 MIN, OK, P/T P-Seals Of 20-3/4"-3K * 13 5/8"-5K C.H.S, NO Success, Try To Remove 2-WAY From T.H.S, OBS. Some Bubble Of LEL Gas, Stop Oper. R/UP Special Lubricator . Remove 2-WAY F/ T.H.S, OBS. The Well For 30 Min, Increase Press. To 27 PSI, Release Press., Open 3SV, Again OBS. The Well For 30 MIN, Increase Press. To 20 PSI, Release Press., Install Slick Line Lubricator, RIH W/ SB Pulling To 2770 M & Remove NO-GO L.N Nipple Prong, POOH To Surface, L/D Prong & SB Pulling, RIH W/ GS Pulling & Remove NO-GO L.N, POOH To Surface. L/D SB Puling & Plug, IN Order To Pass Protection Sleeve Inside The T.H.S, RIH W/ GS Pulling To 30m ,OK, POOH To Surf. In Order To Install Protection Sleeve, RIH W/ Running Tools To 104 M & Install Protection Sleeve Inside The 3SV, POOH To Surface. R/U Coiled Tubing. RIH C.T.B To 400 M, Obs. Increase Well Head Press. To 200 PSI, Close The Well, Monitore Well After 3 Hr, Press: 460 PSI, Open The Well, While Flow The Well, Cont. RIH Coiled Tubing To 4280m Displace Comp. Fluid In Well W/ 240 BBL Diesel, Pull Out Coiled Tubing To Surf. Cont. Flow The Well W/ Chock Sizes: 54 & 48 & 32/64", Oil Weight: 54.5 LB/FT3, BS&W: 0, Ann. Press: 750 PSI, In Order To Monitor The Well & Ann. Press., Close The WELL @ 24:00, While Test Vent Port. Cont. Monitore The Well & Ann. Press., Well Head Press.: 1400 PSI, ANN. PRESS: 0 PSI. Cont. Monitore The Well & Ann. Press., Well Head Press.: 1380 PSI, ANN. PRESS: 0 PSI. Open The Well & Flow It, Choke Size 16/64" (W.H.P: 1271 PSI, ANN. PRESS: 0 PSI) & Choke Size 24/64" (W.H.P: 1160 PSI, ANN. PRESS: 0 PSI) & Choke Size 16/64" (W.H.P: 1271 PSI, ANN. PRESS: 0 PSI, T: 91 F, OIL RATE: 780 BBL/DAY, GAS RATE: 385000 CU.FT/DAY, MW: 54.5). Cont. Flow The Well W/ Chock Size 32 & 36/64" While Test Vent Port, Not OK, Cont. Flow The Well W/ CHOCK Size 32 & 36/64" (WELL HEAD FLOWING PRESS: 1086 PSI, ANN. PRESS: 20 PSI, OIL WEIGHT: 54.5 LB/CU.FT, T: 117 F). R/D C/T Raiser, R/UP Wire Line Lubricator & RIH W/ SB Pulling & Remove Protection Sleeve F/ 3SV, Pull Out To Surf. In Order To Pack The M.Seal, Fasten BMV N&B Again, R/U H-Lubricator & Install 2-WAY In TBG HGR, L/UP Pupm Truck Line To X-MASS TREE , Increase Press. Step By Step To 5500 Psi, W/Out Obs. Any Leakage F/ Vent Port, Release The Press., Try To Remove 2-WAY, Test 3SV For 15 Min, OK, Flow The Well W/ Chock Size 36 & 48" W/ Well Flowing 980 PSI. R/Up Coiled Tubing Raiser Test 3SV For 15 Min, Finally Ok, R/Up Slick Line Lubricator , Again RIH By Slick Line To NO-GO, Install Plug & Prong In NO-GO Landding Nipple, Test W/ 2500 PSI, AFTER 3 TIMES, Finally Ok, Again RIH By Slick Line & Install Protection Sleeve Inside The 3SV. 'R/D Slick Line Lubricator, R/Up Coiled Tubing BOP & Injector, RIH Coiled Tubing To 2770m Displace Oil In Well W/ Diesel, Pull Out Coiled Tubing To Surf., R/UP Slick Line Lubricator & RIH Slick Line To 104m (3SV DEPTH), Remove Protection Sleeve, Pull Out Slick Line & R/D P.S & Lubricator, R/U H-Lubricator Install 2-WAY Inside The Tubing HGR, L/D H-Lubricator, Close Top, Kill, Poduction & Chock Valve Of X-Mass Tree. Release The Rig @ 24:00 on October 29, 2020

1.9 Casing & Liner Depth

Casing Spec.	20"	K – 55, 94#, BTC	0 – 147 m	
	13 – 3/8"	N – 80, 68#, BTC	0 – 1338 m	
	9 – 5/8"	L – 80, 47#, PJE L – 80, 47#, TPCQ	0 – 1765 m	
	7" LNR	L – 80, 29#, TPCQ	1608.2 – 4193 m	
	4-1/2" Non-Slotted Liner	L – 80, 13.5#, BTC	4132.8-4279.6 m	4132.8 – 4199.27 m
	4-1/2" Slotted Liner	L – 80, 13.5#, BTC		4199.27 – 4269.54 m
	4-1/2" Non-Slotted Liner	L – 80, 13.5#, TPCQ		4269.54 – 4279.6 m

Table 7: Casing & Liner Data

1.10 Well Schematic before Workover#1

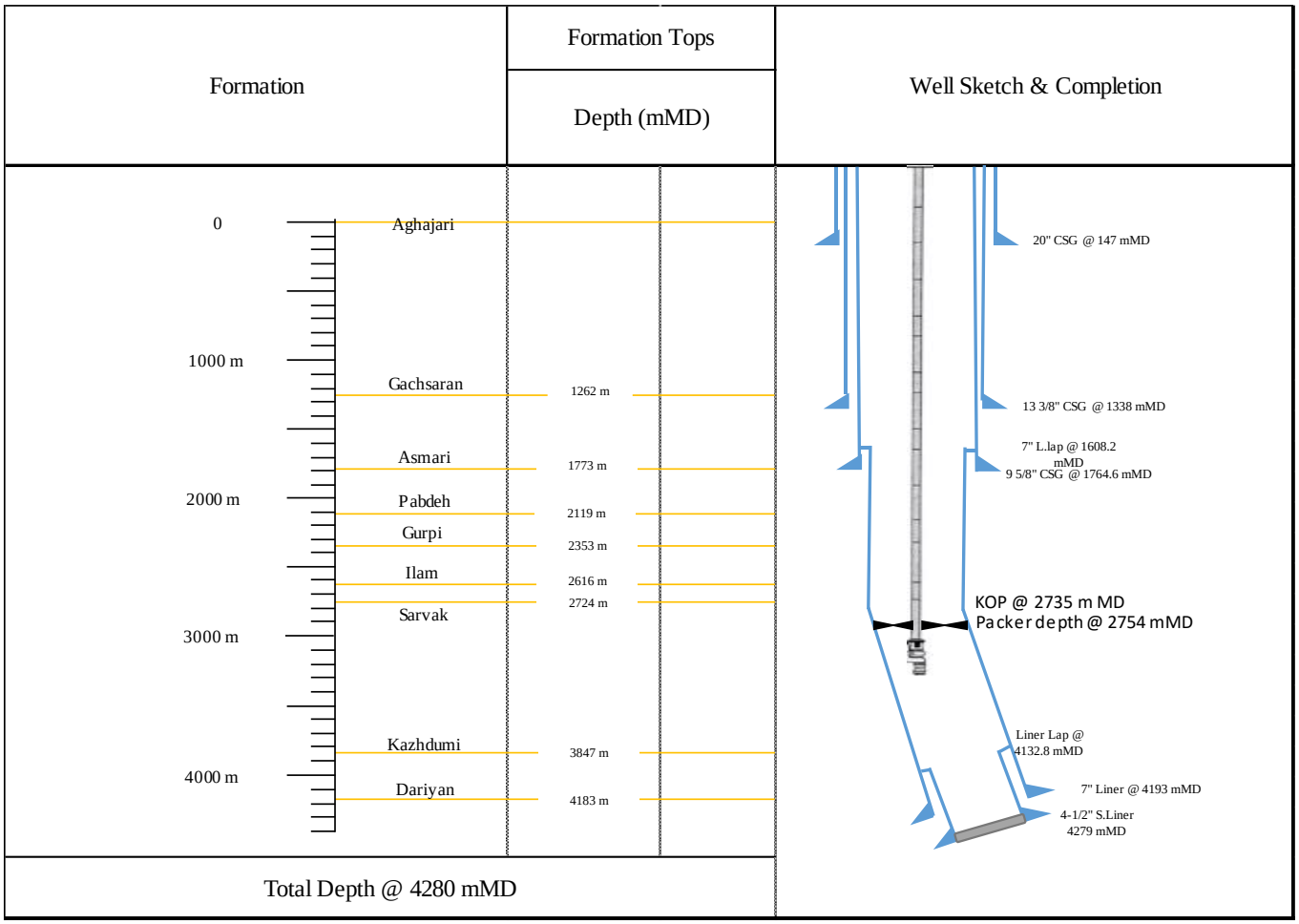


Figure 4: Well Schematic Before workover #1

1.11 Well Schematic after Workover#1

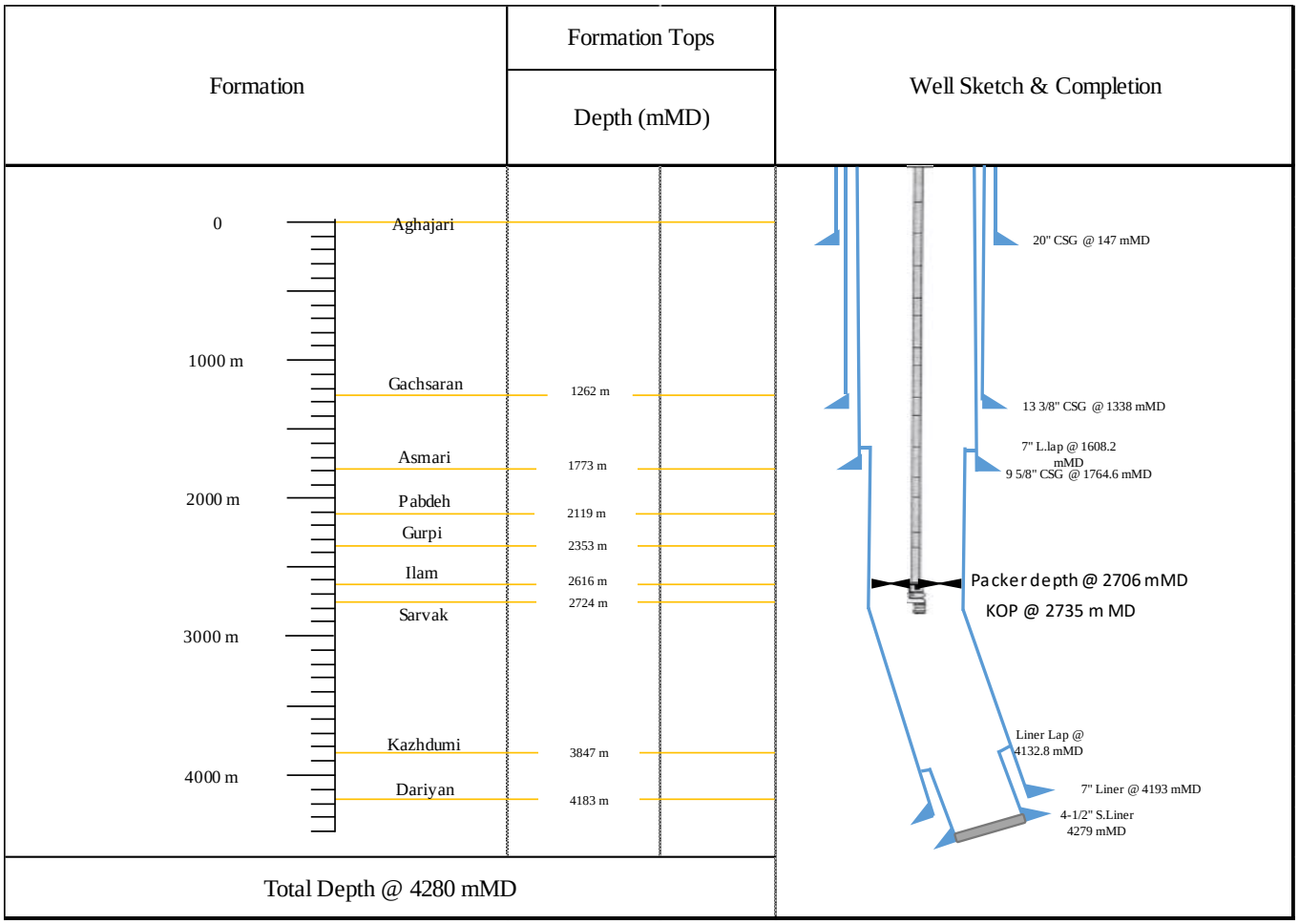


Figure 5: Well Schematic After workover #1

2. Geological Information

2.1 Formation Stratigraphy and Lithology

Stage	Group	Formation	Lithology	Lithology Description	Block I Top (mTVD)	Block II Top (mTVD)	Block III Top (mTVD)
Pliocene Miocene	Fars	Agha Jari		Marl, Sandstone, Anhydrite	Surface	Surface	Surface
Miocene	Fars	Gachsaran		Marl, Anhydrite, Claystone	1380	1150	961
E. Miocene Oligocene		Asmari		Limestone, Sandstone, Claystone	1776	1503	1318
E. Oligocene L. Paleocene		Pabdeh		Limestone, Argillaceous Limestone, Claystone, Dolomite	2079	1786	1525
Maastrichtian Campanian		U. Gurpi Tarbur L. Gurpi		Limestone, Argillaceous Limestone	2352	2303	2153
Santonian Coniacian	Bangestan	Ilam		Limestone, Claystone	2606	2534	2549
	Bangestan	Laffan		Claystone	2691	2642	2625
Turonian Cenomanian	Bangestan	Sarvak		Limestone, Claystone	2697	2648	2632
Cenomanian Albian	Bangestan	Kazhdumi		Sandstone, Argillaceous Limestone, Claystone, Shale	3335	3300	3269
Aptian	Khami	Dariyan		Limestone	3538	3499	3471
Aptian Barremian	Khami	Gadvan		Argillaceous Limestone, Claystone, Sandstone	3718	3696	3635
Neocomian	Khami	Fahliyan		Limestone, Marl	4092	3989	3974

Figure 7: Azadegan Stratigraphic Column

3. Logging Program

Hole Size	Casing Size	Interval (Top-Bottom)	Logs
			Cased Hole
12 1/4"	9 5/8" Casing	0-1608 m	Corrosion log
8-1/2"	7" LNR	1608-4193 m	Corrosion log

Table 8: Logging Program

4. Program Outline

4.1 Contractors Involved

Table 9: Contractor Involved

Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC 201
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	Pars Hafaran Arvand
4	Electrical Logging and Perforation	SEALAND
5	Mud Logging	Pars Geo Data
6	H2S Monitoring and Control	MAPSA KISH BONYAN IMEN
7	Casing and Tubing Running	OEOC
8	Cementing	SPAD ENERGY
9	Dry Test, DST and RTTS	OEOC
10	Slick Line	OEOC
11	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
12	Surface Well Testing	OEOC
13	Liner Hanger	PARS POOLAD
14	Well Head	OEOC

4.2 Time Breakdown

Work-Over Operation TBD of Well: AZNS-166 Work-over (Kazhdumi Producer)	IADC Code	Interval (meters)	Formation	Depth (meters)	ROP	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Work-over												
Check the wellhead for any damage or leakage. Record well head pressures; check all annulus pressure, Bleed off any annulus pressure to Zero. Note: If wellhead pressure is not Zero after bleed off, Contractor shall inform Client. Remedial plan to be prepared in advance by Operators and will be followed after client's confirmation.	De-Completion		Kzh.	4279.6		Rig Less	0.00	0.0	0.0	0.0	0.00	0.0
Retrieve BPV by lubricator and Observe well head pressure	De-Completion		Kzh.	4279.6		Rig Less	0.00	0.0	0.0	0.0	0.00	0.0
Close XMT valves, (Remove 4 1/16" XMT cap).	De-Completion		Kzh.	4279.6		Rig Less	0.00	0.0	0.0	0.0	0.00	0.0
R/u slick line, inspect and check the possibility of install secondary valve in SSSV. If not possible, proceed to operation by Rig	De-Completion		Kzh.	4279.6		Rig Less	0.00	0.0	0.0	0.0	0.00	0.0
Run lock open Tools and punch SSSV install Secondary valve and Test it. If not Successful, proceed to operation by Rig	De-Completion		Kzh.	4279.6		Rig Less	0.00	0.0	0.0	0.0	0.00	0.0
Retrieve the secondary valve then Run gauge ring to top of PLUG and retrieve Plug Note:if observe any obstruction Follow Contingency Plan#01 if successful, proceed to item 33	De-Completion		Kzh.	4279.6		Rig Less	0.00	0.0	0.0	0.0	0.00	0.0
If no success, try to retrieve by coiled tubing If successful, proceed to item 33	De-Completion		Kzh.	4279.6		Rig Less	0.00	0.0	0.0	0.0	0.00	0.0
Held PJSM. Spud the Rig.	De-Completion		Kzh.	4279.6		0	0.00	0.0	0.0	0.0	0.00	0.0
R/U SL,RIH & Try to open SSD,POOH to Surface.Note: if unable to open SSD Follow Contingency Plan#02	De-Completion		Kzh.	4279.6		24	1.00	1.0	1.1	26.4	1.10	1.1
kill the well with proper Mud weight	De-Completion		Kzh.	4279.6		24	1.00	2.0	1.1	26.4	1.10	2.2
Install TWCV in Tubing Hanger. Note:Suitable TWCV Shall be Provided by Client. Note:if Unable to set TWCV Due to Damaged Profile, Install 3SV Blanking plug or Wire Line Bridge Plug	De-Completion		Kzh.	4279.6		24	1.00	3.0	1.1	26.4	1.10	3.3
N/D BMV, N/U BOP, and perform BOP pressure test.	De-Completion		Kzh.	4279.6		60	2.50	5.5	2.8	66.0	2.75	6.1

RIH with TBG hanger retrieving tool and engage with TBG hanger. Note:if Tubing Hanger Thread damaged follow Contingency Plan#03	De-Completion		Kzh.	4279.6		12	0.50	6.0	0.6	13.2	0.55	6.6
Pull Comp. string, Displace completion fluid & oil in hole with proper kill mud and pull out completion string precautionary due to probable trapped pressure below plug & prong meanwhile POOH monitoring the well and annulus to make sure the well is safe and over balance. (from 2754m). Note:if unable to retrieve completion string due to string pack off with huge amount of debries follow contingency plan#04 Note:if unable to retrieve completion string due to string wash out follow contingency plan#05 Note: if observe any damage on THS, follow contingency plan#6 Note:if string stuck while Recovering Completion Packer follow contingency plan#07	De-Completion		Kzh.	4279.6		130	5.42	11.4	6.0	143.0	5.96	12.6
RIH with 8 ½" bit and clean out BHA (P/U DP's JT by JT) to 7" L.LAP (1608.2m), CBU, POOH.	Trip		Kzh.	4279.6		48	2.00	13.4	2.2	52.8	2.20	14.8
RIH with 5-7/8 Mill and clean out BHA to 4-1/2" Slotted Liner Lap @ 4132.8 m, CBU,POOH to Surface.	Trip		Kzh.	4279.6		96	4.00	17.4	4.4	105.6	4.40	19.2
RIH with 3-1/2" Mill and clean out BHA to BTM, take care inside Slotted interval, CBU till shakers clean, POOH. Note:for Corrosion Logging Inside 7" Liner follow contingency plan#8	Trip		Kzh.	4279.6		144	6.00	23.4	6.6	158.4	6.60	25.8
M/U Tandem scraper BHA, RIH with tandem scrapers including 9-5/8" casing scraper & 7" liner scraper to 4130m. Scrap 9-5/8" casing and 7" liner and L/D same.	Trip		Kzh.	4279.6		0	0.00	23.4	0.0	0.0	0.00	25.8
R/U TLC Logging, RIH & Perform Logging (Corrosion Logging) Inside 7" Liner & 9-5/8" Casing,POOH to Surafce.	Log		Kzh.	4279.6		0	0.00	23.4	0.0	0.0	0.00	25.8
RIH scraper and scrape 7" production packer setting area, displace the well with completion fluid, POOH to surface.	Trip		Kzh.	4279.6		72	3.00	26.4	3.3	79.2	3.30	29.1
RIH w/ DP's and DC's, POOH & L/D same.	Trip		Kzh.	4279.6		0	0.00	26.4	0.0	0.0	0.00	29.1
Run 7.5K Kazhdumi completion string.	Completion		Kzh.	4279.6		216	9.00	35.4	9.9	237.6	9.90	39.0
Set TWCV inside TBH, N/D BOP's & install new XMT.	Completion		Kzh.	4279.6		48	2.00	37.4	2.2	52.8	2.20	41.2
Pressure test the XMT.	Completion		Kzh.	4279.6		18	0.75	38.2	0.8	19.8	0.83	42.0
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and Release the Rig.	Completion		Kzh.	4279.6		8	0.33	38.5	0.4	8.8	0.37	42.4
Retrieve TWCV,R/U Slick Line, Set protection sleeve, retrieve plug & prong from NO GO landing nipple, POOH.	Completion		Kzh.	4279.6		Rig Less	0.00	38.5	0.0	0.0	0.00	42.4
RIH with CT to bottom, displace the well with diesel or nitrogen & observ well, POOH.	Completion		Kzh.	4279.6		Rig Less	0.00	38.5	0.0	0.0	0.00	42.4
R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.	Completion		Kzh.	4279.6		Rig Less	0.00	38.5	0.0	0.0	0.00	42.4
Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.	Completion		Kzh.	4279.6		Rig Less	0.00	38.5	0.0	0.0	0.00	42.4
R/U & RIH S.L, Set protection sleeve, POOH & R/D S.L RIH with C/T & perform acid wash as per program, POOH to surface. RIH w/SL and retrieve protection sleeve	Completion		Kzh.	4279.6		Rig Less	0.00	38.5	0.0	0.0	0.00	42.4
Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs. Note 1: The time is only for flowing duration (R/U, R/D & preparation time is excluded). Note 2: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.	Completion		Kzh.	4279.6		Rig Less	0.00	38.5	0.0	0.0	0.00	42.4
RIH with S/L, set protection Sleeve, POOH & R/D SL.	Completion		Kzh.	4279.6		Rig Less	0.00	38.5	0.0	0.0	0.00	42.4

RIH with C/T to bottom & displace the well with diesel, POOH.	Completion		Kzh.	4279.6		Rig Less	0.00	38.5	0.0	0.0	0.00	42.4
RIH with S/L, install plug & prong into NO GO landing nipple, POOH.	Completion		Kzh.	4279.6		Rig Less	0.00	38.5	0.0	0.0	0.00	42.4
RIH with S/L, retrieve protection sleeve.	Completion		Kzh.	4279.6		Rig Less	0.00	38.5	0.0	0.0	0.00	42.4
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.	Completion		Kzh.	4279.6		Rig Less	0.00	38.5	0.0	0.0	0.00	42.4
Total Duration :							***38.5***	***38.5***			***42.4***	***42.4***
NOTE:												
Total Duration for Cut & Slip DL in all hole section.	Cut & Slip			3763		20	0.83	39.3	0.9	22.0	0.92	43.3
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service			3763		21	0.88	40.2	1.0	23.3	0.97	44.2
Total Duration of AZNS-166 Work-over Well:								40.2				44.2
Contingency Plans TBD of Well: AZNS-166 Work-over	IADC Code	Interval (meters)	Formation	Depth (meters)	ROP	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Based on operation situation in each step, the following contingency plans might be followed												
Contingency plan #1: if observe any obstruction while each run w/ Gauge Cutter RIH w/CT and wash from obstruction depth to top of fish with Zylyne.	De-Completion		Kzh.	4279.6		120	5.00	5.00	5.5	132.0	5.50	5.5
Contingency Plan#02 : R/U & RIH W/Coliding Tools & Colide Completion String from Proper Depth,POOH & L/D, Displace well with heavy mud and try to kill the well,Circulate & Balance MW.	De-Completion		Kzh.	4279.6		240	10.00	15.00	11.0	264.0	11.00	16.5
Contingency Plan#03 : RIH w/Taper Tap & Try to tap in & engage w/Tubing Hanger	De-Completion		Kzh.	4279.6		48	2.00	17.00	2.2	52.8	2.20	18.7
Contingency Plan#04 : R/U & RIH with FPIT tools to check string free point,POOH to Surface & L/D.R/U & RIH W/Coliding Tools & Colide Completion String from Proper Depth,POOH & L/D.POOH Completion String from Coliding Depths in Several Runs.	Completion		Kzh.	4279.6		288	12.00	29.00	13.2	316.8	13.20	31.9
Contingency Plan#05 : Pull Completion String untill string will be part from weaked point then POOH & L/D String in Several attempt. It is assumed to part from 4- 1/2" expansion joint as weak point. Mill the remainder of fish & packer. Follow the rest of main plan.	Completion		Kzh.	4279.6		560	23.33	52.33	25.7	616.0	25.67	57.6
contingency plan#06 : if observe any leakage and or failure on THS, secure the well by barriers (RTTS / CEMENT PLUG).	Trip		Kzh.	4279.6		240	10.00	62.33	11.0	264.0	11.00	68.6
contingency plan#07: RIH w/5-7/8" or 6" Mill & Mill the Packer & Tail in Several Runs, POOH to Surafce & L/D mill.	Trip		Kzh.	4279.6		240	10.00	72.33	11.0	264.0	11.00	79.6
contingency plan#08 : Perform Scraper Trip , RIH & Perform Logging (Corrosion Logging) Inside 7" Liner & 9-5/8" Casing,POOH to Surafce.Stand Back required DPs, HWDPs and DC.	Trip		Kzh.	4279.6		336	14.00	86.33	15.4	369.6	15.40	95.0
Total Duration of Contingency Plan AZNS-166 Work-over Well:								86.3				95.0

5. Sequence of Operation

5.1 Main Objective

Decision has been made to de-complete the well & re-complete with new completion equipment.

5.2 Main plan Rig Less Operation

1. Check the wellhead for any damage or leakage. Record well head pressures; check all annulus pressure, Bleed off any annulus pressure to Zero.

Note: If wellhead pressure is not Zero after bleed off, Contractor shall inform Client. Remedial plan to be prepared in advance by Operators and will be followed after client's confirmation.

2. Retrieve BPV by lubricator and Observe well head pressure.
3. Close XMT valves, (Remove 4 1/16" XMT cap).
4. R/u slick line, inspect and check the possibility of install secondary valve in SSSV.
   If not possible, proceed to operation by Rig.
5. Run lock open Tools and punch SSSV install Secondary valve and Test it.
   If not Successful, proceed to operation by Rig
6. Retrieve the secondary valve then Run gauge ring to top of PLUG and retrieve Plug.
   Note: if observe any obstruction Follow Contingency Plan#01
   If successful, proceed to item 29
7. If no success, try to retrieve by coiled tubing.
   If successful, proceed to item 29

5.3 Main plan Operation in Presence of Drilling Rig

8. Held PJSM. Spud the Rig as per PSSR.
9. R/U SL, RIH & Try to open SSD, POOH to Surface.
   Note: if unable to open SSD Follow Contingency Plan#02
10. Kill the well with proper Mud weight.
11. Install TWCV in Tubing Hanger.
    Note: Suitable TWCV Shall be provided by Client.
    Note: if Unable to set TWCV Due to Damaged Profile, install 3SV Blanking plug or Wire Line Bridge Plug
12. N/D BMV, N/U BOP, and perform BOP pressure test.
13. RIH with TBG hanger retrieving tool and engage with TBG hanger.
    Note: if Tubing Hanger Thread damaged follow Contingency Plan#03

14. Pull completion string, displace completion fluid & oil in hole with proper kill mud and pull out completion string precautionary due to probable trapped pressure below plug & prong meanwhile POOH monitoring the well and annulus to make sure the well is safe and over balance. (From 2754m).

Note: if unable to retrieve completion string due to string pack off with huge amount of debris follow contingency plan#04

Note: if unable to retrieve completion string due to string wash out follow contingency plan#05

Note: if observe any damage on THS, follow contingency plan#6

Note: if string stuck while recovering Completion Packer follow contingency plan#07

15. RIH with 8 ½" bit and clean out BHA (P/U DP's JT by JT) to 7" L.LAP (1608.2m), CBU, POOH.

16. RIH with 5-7/8 Mill and clean out BHA to 4-1/2" Slotted Liner Lap @ 4132.8 m, CBU, POOH to Surface.

17. RIH with 3-1/2" Mill and clean out BHA to BTM, take care inside Slotted interval, CBU till shakers clean, POOH.

Note: for Corrosion Logging Inside 7" Liner & 9-5/8" Casing follow contingency plan#8

18. M/U Tandem scraper BHA, RIH with tandem scrapers including 9-5/8" casing scraper & 7" liner scraper to 4130m. Scrap 9-5/8" casing and 7" liner and L/D same.

19. R/U TLC Logging, RIH & Perform Logging (Corrosion Logging) Inside 7" Liner & 9-5/8" Casing, POOH to Surface.

20. RIH scraper and scrape 7" production packer setting area, displace the well with completion fluid, POOH to surface.

21. RIH w/ DP's and DC's, POOH & L/D same.

22. Run 7.5K Kazhdumi completion string.

23. Install TWCV inside TBH, N/D BOP's & install new XMT.

24. Pressure test the XMT.

25. Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and Release the Rig.

Rig less Sequence of Operation

26. Retrieve BPV, R/U Slick Line, set protection sleeve, retrieve plug & prong from NO GO landing nipple, POOH.

27. RIH with CT to bottom, displace the well with diesel & observe well, POOH.

28. R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL. RIH with C/T & perform acid wash as per program, POOH to surface. RIH w/SL and retrieve protection sleeve

29. Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.

30. R/U & RIH S.L, set protection sleeve, POOH & R/D S.L, RIH with C/T & perform acid wash as per program, POOH to surface. RIH w/SL and retrieve protection sleeve.

31. Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs.

Note 1: The time is only for flowing duration (R/U, R/D & preparation time is excluded).

Note 2: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.

32. RIH with S/L, set protection Sleeve, POOH & R/D SL.

33. RIH with S/L, install plug & prong into NO GO landing nipple, POOH.

34. RIH with C/T to bottom & displace the well with diesel, POOH.

35. RIH with S/L, retrieve protection sleeve.

36. Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.

Contingency Plans of Well AZNS-166 Work-over:

Based on operation situation in each step, the following contingency plans might be followed:

Contingency plan #1: if observe any obstruction while each run w/ Gauge Cutter RIH w/CT and wash from obstruction depth to top of fish with Zylyne.

Contingency plan #2: R/U & RIH W/Colliding Tools & Collide Completion String from Proper Depth, POOH & L/D, displace well with heavy mud and try to kill the well, Circulate & Balance MW.

Contingency plan #3: RIH w/Taper Tap & Try to tap in & engage w/Tubing Hanger.

Contingency Plan#04: R/U & RIH with FPIT tools to check string free point, POOH to Surface & L/D. R/U & RIH W/Colliding Tools & Collide Completion String from Proper Depth, POOH & L/D.POOH Completion String from Colliding Depths in Several Runs.

Contingency Plan#05 Pull Completion String until string will be part from weaker point then POOH & L/D String in Several attempt. It is assumed to part from 4- 1/2" expansion joint as weak point. Mill the remainder of fish & packer. Follow the rest of main plan.

Contingency plan#06: if observe any leakage and or failure on THS, secure the well by barriers (RTTS / CEMENT PLUG).

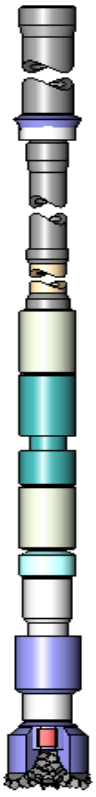
Contingency Plan#07 RIH w/5-7/8" or 6" Mill & Mill the Packer & Tail in Several Runs, POOH to surface & L/D mill.

Contingency plan#08 Perform Scraper Trip, RIH & Perform Logging (Corrosion Logging) Inside 7" Liner & 9-5/8" Casing, POOH to Surface. Stand back required DPs, HWDPs and DC.

6. BHA Program

6.1 6" Milling BHA

BIT SIZE (inch)	6"	BHA Configuration to Milling 7" PKR
Depth In/Out (m)		
HOLE INCLINATION	0	
MUD WEIGHT (pcf)	82	
B.FACTOR	0.833	
Weight in Air (lb)	31,685	
Bouyed Weight (lb)	26,382	
Bouyed Weight Below Jar (lb)	26,382	

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276		-	
	X.O.S	1				0.79	2293.95
	3-1/2" DP	42	3-1/2	2.125		2000	2293.16
	3 1/2" HWDP	6	3-1/2	2.25		57	293.16
	4-3/4" DC	3	4-3/4	2		29	236.16
	4-3/4" D.Jar	1	4-3/4	2		9	207.16
	4-3/4" DC	21	4-3/4	2		196	198.16
	4-3/4" Bit Sub	1	4-3/4"	2		1.23	2.16
	6" M.M J Mill	1	5-7/8"	1.5		0.93	0.93

7. Mud & Cement Program

Note: Detailed mud & cement program, will be issued in due time as per client approved template.

7.1 Workover Hole Section

Workover section	7" & 4-1/2" Slotted Liner
From: 1608.2m	TO: 4280
Kill the well, POOH of Completion String, Complete the hole with Completion & Completion Fluid	

7.1.1 Overview

Drilling Fluid System	Kill Mud and Completion Fluid
Key products	Workover Fluid , Completion Fluid, Kill Mud
Water type	Drill Water
Solid Control	Shale Shaker
Potential problems	Down Hole losses , gas/oil influx,

Interval Mud Properties						
Depth interval (m)	MW (pcf)	Plastic viscosity (cp)	Yield point (lb/100ft²)	API Fluid Loss (cc/30 Min)	pH	Funnel Viscosity (sec/qt)
Surface – 4280	75 – 83	ALAP	12 – 18	<15	9 – 10	40 – 50
Interval Completion Fluid Properties						
Depth interval (m)	MW (pcf)	Plastic viscosity (cp)	Yield point (lb/100ft²)	API Fluid Loss (cc/30 Min)	pH	Funnel Viscosity (sec/qt)
Surface ~ 4280	83 – 84	ALAP	N/A	NC	9.5	27-30

7.1.2 Mud Discussion & Mixing Procedure

7.1.2.1 Mud type

Kill mud 83 pcf will be used to kill the well and well will be displaced by Completion fluid of CaCl2, 83 pcf. tank by the time of displacing the well by kill mud. (As what the hole dictates, mud properties may be varied by the time).

7.1.2.2 Drilling Fluid Formulation

KILL MUD:

75-83 pcf - Salt Saturated / Polymer mud		
Product N.	**Ppb**	**Product Function**
Caustic soda	0.5-1	Alkalinity Control
Soda Ash	1	Hardness Remover
Lime	1	Alkalinity Control
Salt	109	non - solid weighting agent
Starch LV	5 - 6	Fluid loss controller
XC-Polymer	0.5-1	Viscosifier
Limestone	Up to 89	weighting agent
Biocide	As per Requirement	Anti-bacteria

MIXING PROCEDURE

1. Fill the tanks with calculated amount of drill water.
2. Add 109 ppb of NaCl to reach to saturation point at 75 pcf.
4. Raise the pH to 9 – 10 with addition of Caustic soda and Adjust calcium by Soda ash.
5. Add 5-6 ppb of Starch LV at rate of 5 min/Sack.
6. Add calculated amount of Limestone Powder to reach desire MW
7. In case of foaming add required amount of Defoamer.
8. Recheck the pH and use Lime for pH adjustment.
9. Circulate the mud through gun line/Shearing unit at all the time to prevent any settling.
10- Biocide may be used as anti-bacteria agent to prevent polymer degradation.
11. Run a full test on mud and report the properties to office.

DRILLING FLUID SYSTEM	Salt Saturated / Polymer Mud
Key products	Salt (NaCl) , XC-Polymer, Caustic soda , Soda ash, Limestone, Biocide
Water type	Drill Water
Solid Control	Shale shakers , Desilters , Desander, Centrifuge

RQUIRED PROPERTIES:

Fluid Properties Summary		
MUD PARAMETERS		Interval
Density	Pcf	75 - 83
Marsh Viscosity	Sec	40 - 50
PV	Cps	ALAP
Yield Point	lb/100ft²	12 – 18
API filter loss	cc/30min	<15
pH	-	9 – 10
Ca++	mg/l	< 1200
Total Hardness	mg/l	< 1600
LGS	%	< 10
CL^-	Mg/l	18000-188,000

Note: The mud weight and volume of drilling fluid are adjusted in accordance with the prevailing conditions of the well.

MUD VOLUME:

DRILLING FLUID TYPE					
Mud Density (pcf)	75	- 83	LGS Below (%vol)	6	
Bit Size (inch)	0		Solid Control Equipment Eff. %	80	
Section TD (m)	4279		Hole Wash Out %		
Casing OD/ID (inch)	9.625 / 8.835		Dilution Factor (bbl/m)		
Water Depth (m)			Meters To Drilled (m)		

Volume information (Salt Pol Mud)		
Surface System	500	bbl
Cased Hole Volume	725	bbl
New Hole Volume	0	bbl
Wash Out Volume	0	bbl
Sub Total	1,225	bbl
Ex. Vol.	200	bbl
Sub Total	1,425	bbl
Carried from last section	0	bbl
Volume Received	0	bbl
Sub Total	1,425	bbl
New Vol Required	1,425	bbl
Vol transfer out	0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
Soda Ash	110	1	13
Caustic Soda	55	0.5-1	26
Salt (NaCl)	2,200	109	71
Starch LV	55	5-6	155
Limestone Powder	2,200	up to 89	58
Xanthan Gum	55	0.5-1	26
Biocide	55	As Requirement	

Contingency Materials			
Product Name	Function	Unit Size	QTY
Barite	weighting agent	1.5 MT	40
De foamer	De foamer	55 Gal	2
Water Base Mud lubricant	Torque & Drag Reducer/ROP Enhancer	55 Gal	40
Pipe Lax	Pipe freeing agent	55 Gal	2
Citric Acid	Alkalinity control	55 lb	20
Sodium Bicarbonate	CMT Contaminiation Remover	55 lb	20

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.

2-All the calculation based on max material Concentration which mentioned.

COMPLETION FLUID SECTION:

The completion fluid for this section will be fresh water treated with a scavenger, inhibitor and biocide.

SOLIDS – FREE $CACL_2$ FLUID SYSTEM:

This fluid is recommended to make from fresh water or is solutions made with water and the addition of one or more electrolytes. These fluids do not contain either suspended particles or viscosifiers, which mean neither filtration control nor viscosity control, are provided. Some Brine is corrosive and when corrosion inhibitors are required, compatibility studies should be performed. Some brine is toxic and precautions should be taken to prevent contact with unprotected skin. Brine solutions can be prepared to give good inhibition of clays or shales in producing formations. In addition, common contaminants do not affect the brines.

Salt solutions were formerly considered too corrosive to be used as work-over or completion fluids or as packer fluids. The corrosion problem can be solved by adjusting the pH of the solution to above 9 (min) to 10; however, pH measurements may be less important than measuring and controlling the p-alkalinity of the filtrate; this value should be kept above 1 cc. Lime be used for pH control.

Adjustment of pH, or alkalinity, does not necessarily ensure the solution to be corrosion resistant. It may be necessary to use a corrosion inhibitor to achieve lower corrosion rates. In some cases, an oxygen scavenger might be used in conjunction with a filming amine inhibitor.

CORROSION CONTROL:

Corrosion control in completion fluids is important to protect tubular goods and equipment. Corrosion can be limited by controlling the fluid pH at the neutral or alkaline level, and treating out dissolved oxygen. In severe environments such as high temperature or contamination with acid gases, the addition of filming amines can limit corrosion to acceptable levels.

The completion fluid for this section will be Local water treated with a scavenger, and inhibitor.

Pit management will be a key in the successful completion of this interval and sufficient time for planning and preparation should be allowed.

All valves should be checked for leakage beforehand and defects identified to be reported and rectified by drilling contractor.

The mud pits which are to be used to mix the completion brine, should emptied and cleaned adequately and associated lines flushed with water

The rig pumps are to displace all clean up pills as well as the $CaCl_2$ at intended weight. Prepare spacers and wash pills as per program in the tanks. Ensure that sufficient spacer is prepared to account for any dead tanks volume.

Prepare all inhibited $CaCl_2$ before starting the displacement process. After preparing the brine should be filtered through filtration unit and store in a clean pit before displacement.

FILTERATION PROCESS:

Filtration provides clean fluids that will prevent damage to the formation, the reservoir and its permeability. It does this by removal of solids and particulates from the fluid that may otherwise block the open space within the rock formation containing the hydrocarbon fluid within the production zone.

Any solids and contaminates in the completion fluid will reduce its effectiveness. An efficiently filtered and completely solid free completion fluid will add value to both the productivity and the dependability of the well in the long term.

VIRA dual pod filtration cartridge unit is recommended for filtering brines and produced water in completion, stimulation, and workover operations. The pods are connected by a manifold, providing staged filtration capability. Staged filtration utilizes cartridge filters with different ratings to maximize filtration efficiency. Cartridge filtration is designed for use with fluids containing a low percent by volume of suspended solids. Cartridge filtration is not economic when fluids are contaminated with oil or other substances.

Total Solids can be measured at the wellsite using a turbidity meter. Solids that are not added to the system to enhance the performance of a brine are considered contamination. Contaminations include formation clays, precipitates, and polymer residues, among other things. These contaminations can be filtered at the well site using diatomaceous earth, a plate and frame press, and Five-micron to ten absolute cartridges.

Note: a clear completion fluid should not be sent to the wellsite with an NTU (Nephelometric Turbidity Unit) greater than 30 mg/l or a suspended-solid Concentration greater than 40 ppm and for fluid returning from the well NTU should be less than 150.

DISPLACEMENT PROCESS:

It is recommended to circulate and condition the mud prior to displacement and to spot the open hole section to conditioned mud or a solids-free fluid loss pill. Circulate and condition the mud as needed prior to pumping the clean out pills to optimum displacement properties).

After opening the circulating sub, the pumping sequence is as follows:

No.	Description	Volume BBL	Comment
1	Viscous pill	50	Clean pill
2	Continue pumping Completion Fluid until cleanliness specification has been met. Dump return at surface		

- ✓ **Viscous Spacer as clean pill**

NaCl Brine or Fresh water
Xanthan Gum : Viscosifiers

- Mixing procedure:

1. Fill a tanks with required volume NACL Brine or Fresh water for pump 50 bbl Brine.
2. Add 3-4 lb/bbl Xanthan Gum.

- ✓ **Completion fluid**

$CaCl_2$ Brine at 83-84 pcf.
Raise PH to Above 9.5 with Lime
At the end and before pump to the well
Corrosion: Corrosion inhibitor Packer fluid design
H_2S Scavenger: to eliminate dissolved Hydrogen Sulfide from formation
O_2 Scavenger: Oxygen scavenger (to eliminate dissolved oxygen as source of corrosion)

Mixing procedure for 1 BBL 80 pcf Brine:

1. Fill the tank with 0.8 bbl. fresh water
2. Add 187-200 ppb $CaCl_2$.
3. Raise PH to above 9.5 with 3 - 4 ppb Lime
4. Add 1% V/V Corrosion Inhibitor
5. Add 0.5% V/V H_2S Scavenger
6. Add 0.5% V/V O_2 Scavenger

RIG CLEAN UP CHECK LIST (Should be done by Rig Operator)

A. Areas to clean.

1. Surface pits.
2. Clean the suction, return, settling, and reserve pits.
3. Clean the sand traps.
4. Clean the ditches and troughs.
5. Clean the trip tank.
6. Clean the shaker area including all solids equipment and lines.
7. Clean the mixing hoppers.
8. Clean the cement unit.
9. Clean any other pits or equipment that the completion fluid may contact.
10. Cycle all valves in mixing system.

B. Items to flush out.

1. All mud pumps and suction lines.
2. All centrifuges and lines.
3. Chemical and weighting mixing lines in contact with the completion fluid.
4. Lines from the pit room to the trip tank.
5. Lines from the rig floor to the trip tank.
6. Choke, choke manifold and kill lines.
7. Standpipes.
8. Degassing and solids control equipment.
9. Lines to the cement unit and/or other pumping equipment.
10. Casings fill up line.
11. Lines to and from the filtration line.
12. Take on and return lines to and from the boat.
13. Any other lines that the completion fluid may contact.

DISPLACEMENT PROCEDURE:

Pre-Displacement Steps

1. Run bit and dual or tandem scrapers.
2. Circulate and condition mud. Reduce funnel viscosity as much as possible while maintaining adequate rheological properties to clean hole and prevent settling of solids. Maintain critical annular velocity to assure that liner tops are thoroughly cleaned prior to displacement.

3. Rig up equipment for displacement of completion fluid. All tanks, circulating lines, and pumps to be used in the displacement should be cleared of residual mud.
4. Insure sufficient space is available to store displaced mud.

DRILLING FLUID TYPE					
Mud Density (pcf)	83	-	84	LGS Below (%vol)	
Bit Size (inch)		0		Solid Control Equipment Eff. %	
Section TD (m)		4200		Hole Wash Out %	
Casing OD/ID (inch)				Dilution Factor (bbl/m)	
Water Depth (m)				Meters To Drilled (m)	

Volume information (NaCl Brine)		
Surface System	250	bbl
Cased Hole Volume	761	bbl
New Hole Volume	0	bbl
Wash Out Volume	0	bbl
Sub Total	1,011	bbl
Dilution Volume	0	bbl
Sub Total	1,011	bbl
30 % Extera Vol	303	bbl
Volume Received	0	bbl
Sub Total	1,314	bbl
New Vol Required	1,314	bbl
Vol transfer out	0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
Lime	55	3-4	96
CaCl2	2,200	187-200	119
Corrosion Inhibitor	55 Gal	1% V/V	10
H_2S Scavenger	55 Gal	0.5% V/V	5
O_2 Scavenger	55 Gal	0.5% V/V	5

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.

2-All the calculation based on max material Concentration which mentioned.

7.2 Cement Program

Not applicable.

8. Completion

Completion History:

RIH 3-1/2" Completion String as Program to 2783.7m While Press. Test Comp. String Several Steps W/ 4500 PSI / 15 MIN, OK. cont. RIH Comp. String to 2785.2m, Land TBG HGR in 11"-10K T.H.S, Press. Test Comp. String W/ 4500 PSI / 15 MIN. Run in Comp. String W/ SB Pulling & Plug to 2770.1m Set Prong in NO-GO. Pressure test Comp. String W/4500psi for 15MIN, Ok. Close 3SV, laydown 41/2"L. JT, install 4"TWCV in TBG HGR, P/T TWCV & TBG HGR W/ 4500 Psi, OK, N/D BOP Stack, Install 11"*4 1/16"-10K BTM Master Valve.

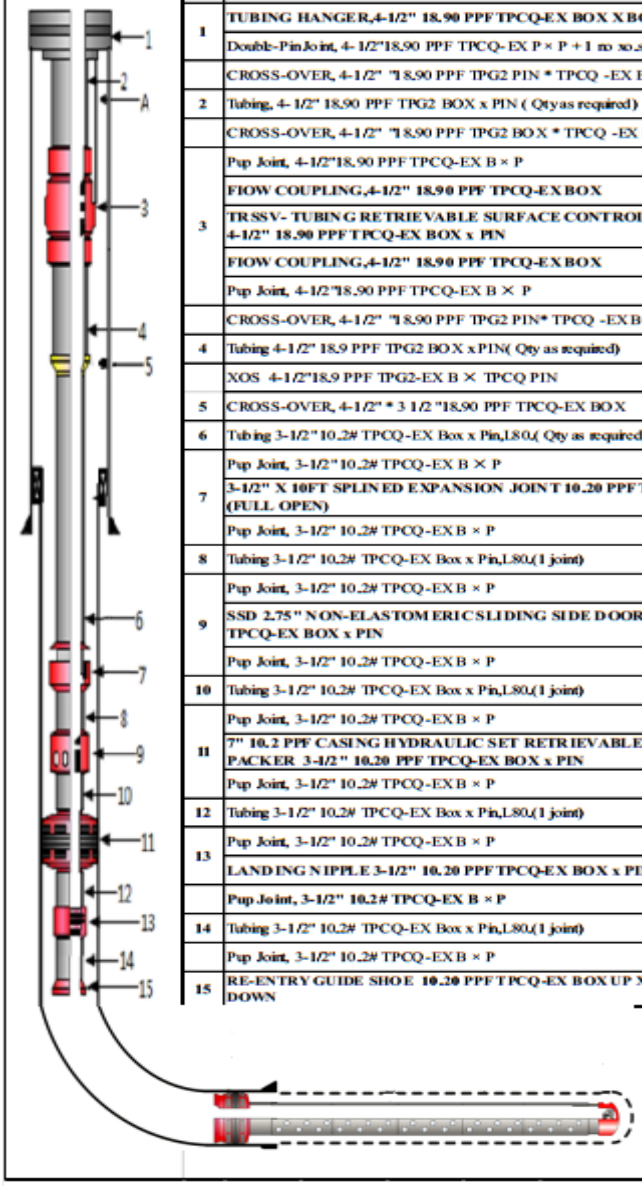
8.1 Equipment Specification

Item	Equipment Description	
1	**CONTROL LINE:**	
	THICKNESS:	0.049"
	WORK PRESSURE:	5000 PSI
	O.D.:	0.250"
	TYPE:	SEAMLESS
	MATERIAL:	316L
	LENGTH:	120m
2	**UPPER FLOW COUPLING:**	
	OD:	5.250"
	ID:	3.640"
	THREADS:	TPCQ-EX
	WORKING PRESSURE:	5000 PSI
	LENGTH:	6 FT
	MATERIAL:	L-80
	LOWER FLOW COUPLING:	
	OD:	5.250"
	ID:	3.640"
	THREADS:	TPCQ-EX
	WORKING PRESSURE:	5000 PSI
	LENGTH:	4FT
	MATERIAL:	L-80
3	**TUBING RETRIEVABL OR WIRELINE SAFETY VALVE**	
	SETTING DEPTH:	105.9m
	WORKING PRESSURE:	5000 PSI
	TEMPERATURE:	250 °F
	MAX O.D.:	6.97"
	MIN. I.D.:	3.437"
	PROFILE:	RQ
	SELF-EQUALIZING FEATURE	YES
	MATERIAL:	9CR1MO
	THREADE:	TPCQ-EX
4	**EXPANSION JOINT**	
	STROKE LENGTH	10 FT
	SHEAR PIN SET:	24,000 LBS (WELL SITE ADJUSTABLE)
	WORKING PRESSURE:	5000 PSI
	TEMPERATURE:	250 DEG F
	TOOL OD:	4.88"
	TOOL ID:	2.901"
	THREADS:	3-1/2" 10.20 PPF TPCQ-EX BOX* PIN
	MATERIAL:	9CR1MO
	TRANSFER TORQUE FEATURE	YES

5	**SLIDING SIDE DOOR**	
	PROFILE:	XN
	MAX.OD:	4.551"
	MIN ID:	2.750"
	PRESSURE RATING:	5000 PSI
	TEMPERATURE:	250 DEG F
	THREADS:	3-1/2" 10.20 PPF TPCQ-EX BOX X PIN
	MATERIAL:	9CR1MO
	OPEN METHOD	DOWNWARD OPEN TOOLS
6	**COMPLETION PACKER**	
	TYPE:	TUBING RETRIEVABLE
	CASING :	7" 29 PPF
	PROFILE:	RP
	OD:	5.945"
	ID:	2.922"
	WORKING PRESSURE:	5000 PSI
	TEMPERATURE:	250 DEG F
	THREADS:	3-1/2" 10.20 PPF TPCQ-EX BOX X PIN
	MATERIAL:	9CR1MO
	ELEMENT MATERIAL:	AFLAS
7	**NO-GO LANDING NIPPLE**	
	PROFILE TYPE:	RN
	WORKING PRESSURE:	5000 PSI
	MAX O.D.:	4.250"
	SEAL BORE:	2.562"
	MIN ID.:	2.442"
	THREADS:	3-1/2" 10.20 PPF TPCQ-EX
	MATERIAL:	9CR1MO
8	**3-1/2" MULE SHOE GUIDE**	
	OD:	4.250"
	MIN ID:	2.900
	TOP CONNECTION	3 ½'' 10.20 PPF TPCQ-EX BOX
	MATERIAL:	9CR1MO

8.2 First Run Completion String Which was Failed (Weqixin Co.)

AZNS-166 (Pad 9), RIG 28 FATH, Well Completion String Schematic Diagram



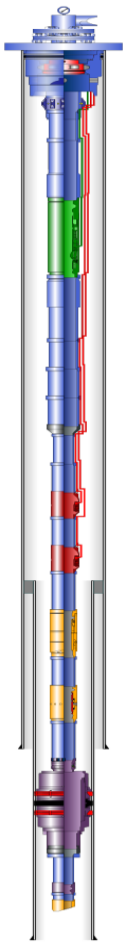
Item	DESCRIPTION	ID IN	OD IN	LENGTH M	TOP DEPTH M	BOTTOM DEPTH M
	RKB			10.210		10.020
1	TUBING HANGER,4-1/2" 18.90 PPF TPCQ-EX BOX X BOX		11.000	0.540	10.020	10.560
	Double-Pin Joint, 4-1/2"18.90 PPF TPCQ-EX P × P + 1 no so.s TPG2 (9.09 M)	3.640	4.500	1.627	10.560	12.187
	CROSS-OVER, 4-1/2" "18.90 PPF TPG2 PIN * TPCQ -EX BOX	3.640	4.500	9.090	12.187	21.277
2	Tubing, 4-1/2" 18.90 PPF TPG2 BOX x PIN (Qty as required)	3.640	4.500	73.871	21.277	95.148
	CROSS-OVER, 4-1/2" "18.90 PPF TPG2 BOX * TPCQ -EX PIN	3.640	4.500	9.207	95.148	104.355
3	Pup Joint, 4-1/2"18.90 PPF TPCQ-EX B × P	3.640	4.500	1.580	104.355	105.935
	FIOW COUPLING,4-1/2" 18.90 PPF TPCQ-EX BOX	3.640	5.250	1.353	105.935	107.288
	TRSSV- TUBING RETRIEVABLE SURFACE CONTROLLED SAFETY VALVE 4-1/2" 18.90 PPF TPCQ-EX BOX x PIN	3.437	6.970	1.740	107.288	109.028
	FIOW COUPLING,4-1/2" 18.90 PPF TPCQ-EX BOX	3.640	5.250	0.899	109.028	109.927
	Pup Joint, 4-1/2"18.90 PPF TPCQ-EX B × P	3.640	4.500	1.608	109.927	111.535
	CROSS-OVER, 4-1/2" "18.90 PPF TPG2 PIN* TPCQ -EX BOX	3.640	4.500	8.979	111.535	120.514
4	Tubing 4-1/2" 18.9 PPF TPG2 BOX x PIN(Qty as required)	3.640	4.500	1433.380	120.514	1553.894
	XOS 4-1/2"18.9 PPF TPG2-EX B × TPCQ PIN	3.640	4.500	9.214	1553.894	1563.108
5	CROSS-OVER, 4-1/2" * 3 1/2 "18.90 PPF TPCQ-EX BOX	2.922	5.250	0.946	1563.108	1564.054
6	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80,(Qty as required)	2.922	3.500	1155.454	1564.054	2719.508
7	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.685	2719.508	2721.193
	3-1/2" X 10FT SPLINED EXPANSION JOINT 10.20 PPF TPCQ-EX BOX x PIN (FULL OPEN)	2.901	4.881	6.261	2721.193	2727.454
	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.489	2727.454	2728.943
8	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80,(1 joint)	2.922	3.500	9.533	2728.943	2738.476
9	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.688	2738.476	2740.164
	SSD 2.75" NON-ELASTOMERIC SLIDING SIDE DOOR 3-1/2" 10.20 PPF TPCQ-EX BOX x PIN	2.750	4.551	1.365	2740.164	2741.529
	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.488	2741.529	2743.017
10	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80,(1 joint)	2.922	3.500	9.541	2743.017	2752.558
11	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.488	2752.558	2754.046
	7" 10.2 PPF CASING HYDRAULIC SET RETRIEVABLE PRODUCTION PACKER 3-1/2" 10.20 PPF TPCQ-EX BOX x PIN	2.880	5.827	3.238	2754.046	2757.284
	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.688	2757.284	2758.972
12	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80,(1 joint)	2.922	3.500	9.531	2758.972	2768.503
13	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.488	2768.503	2769.991
	LANDING NIPPLE 3-1/2" 10.20 PPF TPCQ-EX BOX x PIN (NO-GO ID: 2.635)	2.562	4.251	0.614	2769.991	2770.605
	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.688	2770.605	2772.293
14	Tubing 3-1/2" 10.2# TPCQ-EX Box x Pin,L80,(1 joint)	2.922	3.500	9.595	2772.293	2781.888
	Pup Joint, 3-1/2" 10.2# TPCQ-EX B × P	2.922	3.500	1.601	2781.888	2783.489
15	RE-ENTRY GUIDE SHOE 10.20 PPF TPCQ-EX BOX UP X MULE SHOE GUIDE DOWN	2.901	4.299	0.254	2783.489	2783.743

Figure 9 2nd Completion Schematic

8.3 Proposed Completion string

AZNS-166 Completion Sketch

Contractor		Liner & Tubing Data						Client	
Field	AZADEGAN	Description	7" Liner	4-1/2" Liner	3 1/2" Tubing	4 1/2" Tubing	Perforated Depth	7" Liner Lap	1608.2
Well Number:	AZNS-166	OD[in]	7.00	4.50	3.50	4.50	4199.27 - 4269.54	7" Liner Shoe	4193
Rig :	OEOC-201	Nominal weight [lb/ft]	29.00	13.50	10.20	18.90		4-1/2" Liner Lap	4132.8
RTE-GLE	3.95	Grade	L - 80	L-80	L-80	L-80		4-1/2" Liner Shoe	4279.6
RTE-THF	14.45	Thread	TPCQ	BTC - TPCQ	HSM3	HSM3		Total Depth:	4280
Rig Spudded Date :	TBA	Inside diameter [in]	6.184	3.920	2.92	3.640		X-mass Tree:	MSP 10k
End of Operation	TBA	API Drift Diameter	6.059	3.735	2.80	3.515		Size & Rating:	TBA

Schematic		Description	Max OD	Min ID	Length(m)	BTM(m)	TOP(m)
	1	Elevation (floor - TBG Head Flange)				11.69	0.000
	2	TBG HANGER 11" * 4-1/2" ACME Box up * 4-1/2" TPCQ-EX Box Down , M/C: FF-0.5 T/C: U PSL: 3G PR: 2, 4" Type "H" Two Way	11	3.68	0.6	12.29	11.690
	3	4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ-EX PIN ON TOP X HSM3 PIN ON DOWN* 9Cr1MO	5.5	3.64	1.01	13.300	12.290
	4	5 JOINTs TUBING 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	47.5	60.800	13.300
	5	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.62	62.420	60.800
	6	4-1/2" FLOW COUPLING,18.9#,9Cr1MO,HSM3 , 6 FT	5.5	3.64	1.72	64.140	62.420
	7	TRSSV Tubing Retrievable Saftey Valve, 4-1/2", 18.9# w/ 3.437" Seal bore,HSM3 ,9Cr1MO, 7.5K WP with AF Profile, Flapper Type	7.5	3.437	1.78	65.920	64.140
	8	4-1/2" FLOW COUPLING, 18.9#,9Cr1MO,HSM3 , 4' FT	5.5	3.64	1.11	67.030	65.920
	9	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.62	68.650	67.030
	10	151 JOINTs TUBING 4-1/2"HSM3 18.9#, L-80	5.118	3.64	1434.5	1503.150	68.650
	11	X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	5.25	2.922	1.02	1504.170	1503.150
	12	121 JOINTs TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	1149.5	2653.670	1504.170
	21	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	2655.490	2653.670
	22	3-1/2" EXP. JOINT,HSM3 , 9Cr1MO, 10' STK (Half Closed) Splind Rotational Lock System Any Position, Shear Pin Set 15,000 LBS,	5.315	2.922	5.755	2661.245	2655.490
	23	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.822	2663.067	2661.245
	24	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	2672.567	2663.067
	25	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.625	2674.192	2672.567
	26	3-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, 9Cr1MO, HSM3 with Type X Selective Profile (PROFILE: 2.750" X, DOWNWARD OPEN)	4.551	2.75	1.19	2675.382	2674.192
	27	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.7	2677.082	2675.382
	28	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.27	2686.352	2677.082
	29	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.725	2688.077	2686.352
	31	7" 29 PPF CASING HYDRAULIC SET RETRIEVABLE PRODUCTION PACKER, L-80, HSM-3 Pin & Box, 9cr1Mo, TO BE RUN IN 7.00" 29 PPF CASING	5.910	2.922	4.44	2692.517	2688.077
	33	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.822	2694.339	2692.517
	34	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	2703.839	2694.339
	35	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.825	2705.664	2703.839
	36	3-1/2" BOTTOM NO-GO LANDING,HSM3 , 10.2#, 9Cr1MO w/ Seal Bore 2.75" TYPE XN (NO-GO ID: 2.635")	4.5	2.75	0.37	2706.034	2705.664
	38	RE-ENTERYY MULE SHOE, L-80, HSM-3 Box, 9cr1Mo	4.25	2.922	0.37	2706.404	2706.034
		END OF COMPLETION STRING					2706.404

Tubing Retrievable Saftey Valve	65.920	64.140
Traveling Joint	2661.245	2655.490
SSD	2675.382	2674.192
7" HYDRAULIC SET RETRIEVABLE PRODUCTION PACKER	2692.517	2688.077
NO-GO Landing Nipple	2706.034	2705.664
Entry Guide	2706.404	2706.034

Figure 11: Proposed Completion Schematic

9. BOP, Wellhead & X-Mass Tree

9.1 Existing Wellhead & X-Mass Tree

Solid Block, 4-1/16",10K	
Manufacturer	MSP/DRILEX
APA 6A	
Serial Number	N160006
Material Class	FF-0.5
PSL: 3G	PR: 2
BTM Master Valve, 11",5K x 4-1/16", 10K	
Manufacturer	MSP/DRILEX
Serial Number	008929S160003
Material Class	FF-0.5
API 6A	
PSL/PR	3/2
Tubing Head Spool,13-5/8",5K x 11",10K	
Manufacturer	MSP/DRILEX
Material Class	EE-0.5
API 6A	
Serial Number	TH1621N140003
PSL: 3	PR: 2
Casing Head Spool, 20-3/4",3K x 13-5/8",5K	
Manufacturer	MSP/DRILEX
Material Class	DD-0.5
API 6A-1677	
Serial Number	CS1523N140058
PSL: 1	PR: 1
Casing Head Housing 20 3/4"x 3K-13 5/8"x5, SLIPS TYPE	
Manufacturer	MSP/DRILEX
Material Class	DD-0.5
API 6A	
Serial Number	CH1521N150007
PSL: 1	PR: 1

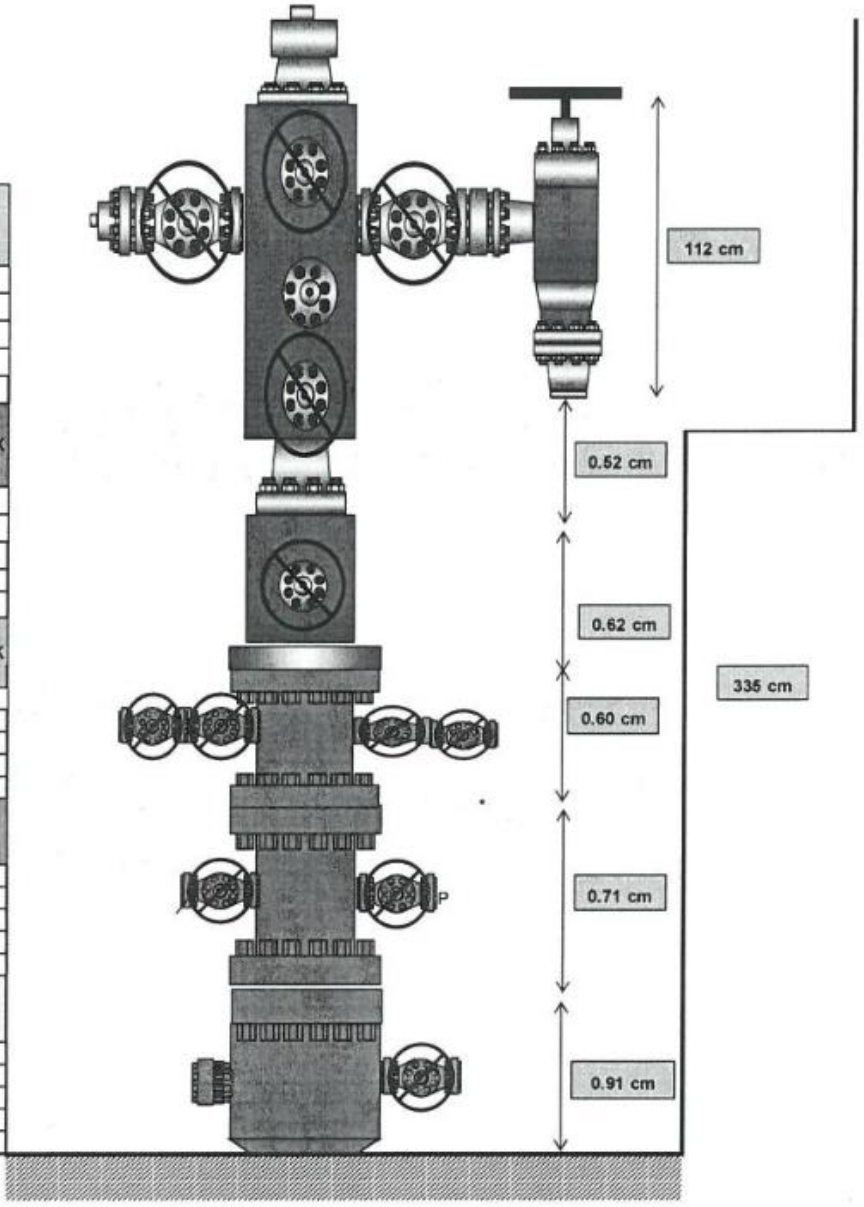
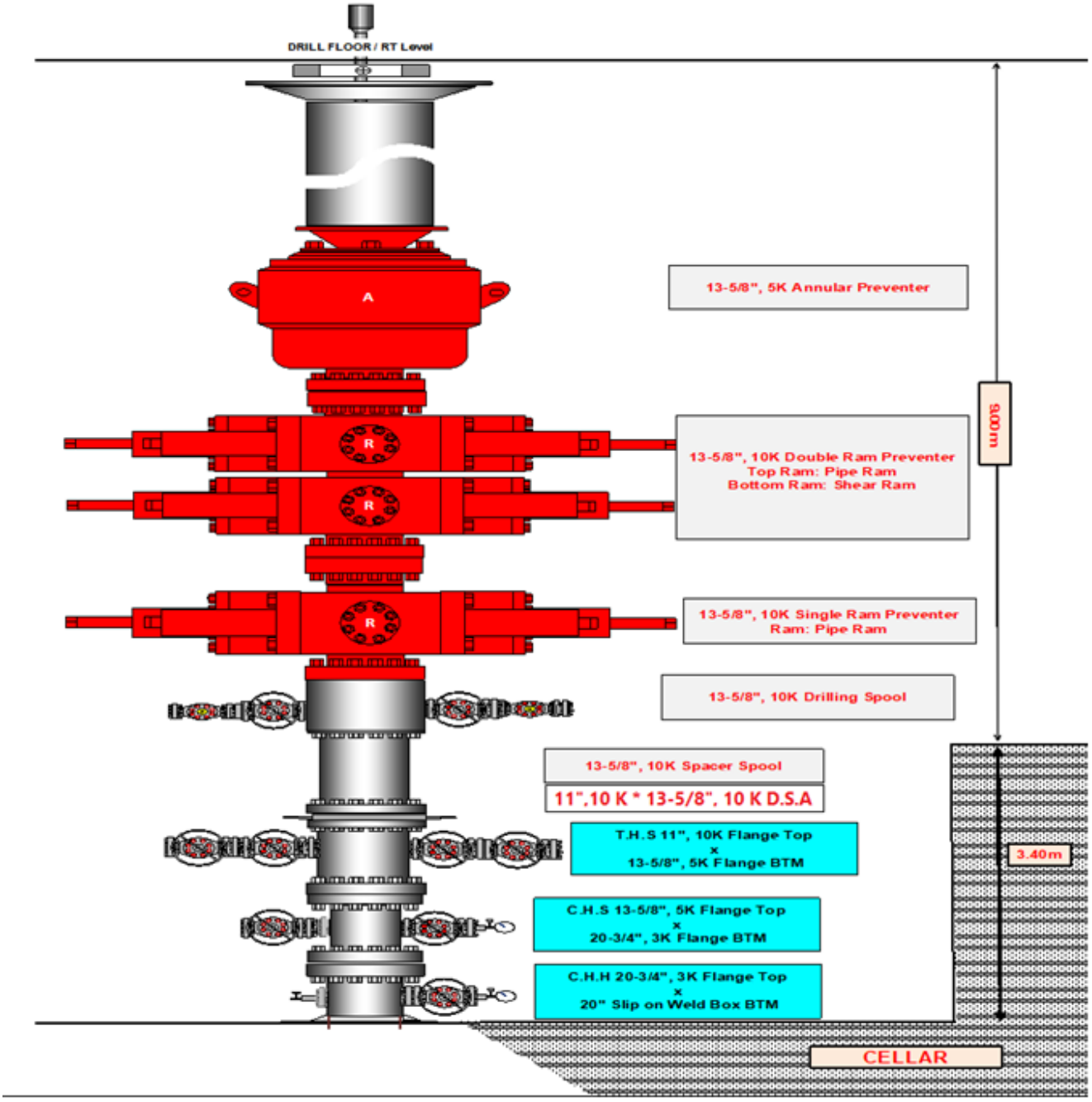


Figure 12: Existing Wellhead

9.2 BOP

Figure 13: BOP Sketch

10.



10. Waste Management

There is no drilling objective and drilling cutting in AZNS-166 workover well; therefore waste management program is not applicable.

In case of requirement to dispose wastewater in workover well, waste management program will be issued at due time as PEDEC policy.

11. Health, Safety and Environments

12.1 WELL CONTROL

All the drilling operation shall obey the requirement of Well Control Manual of MSA and contractor's well control manual; under no circumstances should the deviation from well control policy be permitted without prior approval through MSA. If the contractor's well control manual or anywhere has conflict with Well Control Manual of MSA, take Well Control Manual of MSA as the standard.

12.1.1 WORK Requirement of Well Control Equipment

- With shut-in casing pressure lower than 1,000psi, while snubbing with annular preventer, drill tools shall have 18-deg shoulder tool joints, and the snubbing velocity shall not be greater than 0.2m/s.
- After shut off the annular preventer or ram preventer, with the shut-in casing pressure not greater than 2,000psi, drill tools are allowed to pull up and down with the velocity less than 0.2m/s, however, rotating or tool joint passing through are forbidden.
- While drilling through pay zone and thereafter, BOP function (close-open) test for the BOPs shall be performed regularly.
- Kill manifolds shall not be used for drilling fluid fill-in.
- The maximum allowed shut-in casing pressure shall be remarkably indicated on choke manifolds.
- All gate valves of well control manifolds shall be tagged with serial numbers, and be indicated with on/Off.
- In case shear rams were used, after shearing the drill pipe with shear rams, retrieve the upper sheared drill pipe, lower a drill pipe joint with TIW valve installed into the hole and close pipe rams on it to prevent the possibility of shear ram's failure.

12.1.2 Management of Well Control Equipment

- Well control equipment used by the drilling crew shall be managed and operated by designated personnel with clear post responsibilities.
- Well site shall have specified component storeroom and rubber element storeroom with air conditioners respectively. The temperature of storerooms shall meet storage requirements of components and rubber elements.
- At least 1 set is available as back-up for every type ram assembly in the well site. Ram preventer side door seals and ram rubber, every type gasket and bolt etc. should be enough for backup.

12.2 Recording Reference

All BOP tests, drills, function tests, any malfunctions, repair or maintenance to the mud System and well control equipment shall be recorded in the IADC daily reports and shall be signed by both the Drilling Contractor's Tool Pusher and Drilling Supervisor on the well site. They shall also be recorded in the Daily Drilling Report.

12.3 Flow Check Procedure

To avoid a kick situation, it is recommended that a flow check be conducted at specific times during the drilling operation.
While drilling (any warning sign).
Prior to tripping the drill string.
While tripping (any warning sign).
While out of hole (continuously).
Trip out:
After pulling 5% of the drill pipe
At the midpoint depth of the wellbore
Prior to pulling the first stand of drill collars
After all the drill string is out of hole.
Trip in:
Upon reaching surface casing shoe
At midpoint of the wellbore

12.4 Primary Control Reference

It is the responsibility of the Drilling Contractor to keep the hole full at all times by using all the available equipment. The Drilling Supervisor shall monitor that the correct preventive practices are being implemented and/or conducted at all times.
Extreme care shall be taken to monitor mud volume, drilling breaks and gas cut mud.
The MACP (Maximum allowable casing pressure) shall be confirmed by the Drilling Supervisor, either after setting each new casing string or, whenever the density of the drilling mud changes.
The MACP shall be clearly written on a kick control sheet that will be posted near the choke control panel.
The predetermined distance at which the Tool Joint is to be pulled above the rotary should be posted on rig floor near the BOP control panel.
The Driller is responsible for carefully measuring and recording the slow pump pressure.
Awareness of these values is an important element in killing operations, in order to avoid formation breakdown.

Take a measure and record the slow pump pressure at three different flow rates with all pumps at the following times:
Once per tour.
Whenever changes occur in the mud density/rheology.
Whenever changes occur in the dimension and characteristics of the string, BHA, jet size, jet plugged or jet lost, etc.

12.5 Secondary Control Reference

Any time a drilling break is noticed, drilling shall stop and a static monitoring of the well shall be carried out. Space-out for all Pipe Rams and Annular Preventer must be known to the Driller at all times to keep tool joints clear of annular preventer and rams.

12.6 Well control during drilling and completion operation

The drilling design shall be strictly implemented by the drilling contractor and when it is discovered there is incompliance of downhole situation with design and the design will be amended, the procedure for amendment of design shall be followed. If emergency is encountered, and safe operation can be affected, after the discussion between drilling rig manager and the drilling supervisor, practical technical measures can be taken first and then report to the top management. The Company shall approve and notice in time as per the drilling program amendment procedure.
The curve of well equivalent mud weight as per predicted formation pressure, curve of mud weight in drilling program and curve of actual mud weight must be drawn during drilling operation. For exploratory well and key well, the curve of dc index while drilling formation pressure equivalent mud weight for sandstone, mudstone and shale shall be drawn by the comprehensive mud logging crew.
When entering oil and gas formation or drilling master reservoir, high diameter nozzle shall be selected for the drilling bit to treat well killing and mud loss.
SCR (Slow Circulation Rate) shall be conducted at the start of drilling activity, pump strokes, pumping rate and circulating pressure should be recorded. While drilling in pay zone, SCR is required to be conducted with all pumps separately as follows, pump strokes, pumping rate and circulating pressure shall be recorded in Daily Drilling Report and Kill Sheet.
Once for every 12 hours in general;
After bit change or BHA change; 3) Mud weight change exceeds $^{3.75}$pcf.
Oil and gas invasion and overflow shall be checked with wiper trip and shall be conducted under the following situation:
Each time before POOH after drilling oil and gas formations.
Severe invasion of oil and gas occurs during drilling, but overflow does not occur before POOH.
POOH after killing well.
POOH after lowering mud weight in wellbore.
POOH after curing mud loss during drilling oil and gas formation.
POOH after drilling oil and gas formation and a prolonged period of time is needed for other operations (E logging, running casing, running tubing and DST etc.).
Regarding the shut-in methods selection
Soft shut-in method is recommended for gas well;
Hard shut-in method is recommended for oil well;
Hard shut-in method is recommended for the well of which the well has strong formation strength in EMW as per FIT test;

Soft shut-in method is recommended for the well of which the well has weak formation strength in EMW as per FIT test.

Main technical measures for prevention of kicking and blowout during tripping.

Before POOH, circulation duration of mud in wellbore shall not be less than one and a half cycles. The observation time after wiper trip shall be no less than one and a half cycles. The density differential of inlet and outlet shall not exceed 1.248 pcf.

Flow check shall be conducted regularly via trip tank as follows

Prior to POOH (keep string rotation at 10 rpm for preventing stuck pipe)

POOH to last casing shoe

Before POOH BHA through BOP stack

Note: Tripping should be shut-off immediately and flow check shall be performed later on when any suspected well kick is observed.

During POOH, fill up hole every 3-5 stands of DP and every stand of DC. If the nozzle is plugged, during POOH, mud shall be filled full for each joint or continuous filling shall be conducted and trip sheet shall be recorded carefully.

When the bit is in the oil and gas formation and 300m above top of oil and gas formation, POOH speed shall be controlled to maintain the excellent well building property and rheological property of mud to avoid severe swabbing in the wellbore.

During RIH, the running speed shall be controlled to avoid the loss of circulation due to down hole pressure surge.

During tripping, check and repair of the equipment shall be arranged at the time when running to the casing shoe. if equipment have to be checked and repaired during POOH, due blowout prevention measures shall be taken during and after the check and repair, drill string shall be run to bottom and circulate for one and a half cycles. After everything is normal, drilling shall be resumed. Equipment check and repair shall not be conducted for empty well condition.

Treatment measures for invasion of gas during normal drilling:

When invasion of gas is discovered, it shall be discharged with circulation in time. The gas invaded mud shall not be pumped in the hole before eliminating gas.

Increasing mud weight shall be conducted when stop drilling and after invaded gas eliminated from mud. Mud weight incremental of each circulation cycle shall be controlled within 3.121 pcf

"In-post" policy

"In-post" will be initiated after the surface casing is cemented.

The "in-post" personnel must be trained and verified qualified by the technical staff of drilling rig.

During drilling, specific personnel shall be arranged to be "in-post", the mud logging personnel shall monitor the level of the circulating tank etc. as per the requirements. During tripping, other auxiliary operation or drilling suspension, specific personnel for "in-post" shall be arranged by the drilling and mud logging personnel in the meantime.

"In-post" requirements:

For drilling operation, variation of the outlet flow, drilling time, lithology, bubble, smell, showings, pit volume, density and mud viscosity, gas measured values and content of chloride etc. shall be carefully monitored. The level of the circulating tank shall be observed and recorded each 15 minutes and if special situation occurs, the measurement and record shall be intensified. If anomaly occurs, it shall be reported to the driller in time.

For tripping operation, trip sheet shall be filled accordingly. If anomaly occurs, it shall be reported to the driller.

For other operations, it shall be observed whether there is overflow of mud at the wellhead or mud level from wellhead. If anomaly occurs, it shall be reported to the driller.

Treatment of loss of circulation

Before cementing non-target zone or pay zone, if loss of circulation occurs during drilling, POOH shall be carried to run open ended drill pipe for cement plug job if condition allows. During RIH, mud shall be filled from annulus to balance formation pressure to avoid blowout.

Treatment of loss of circulation during drilling after production casing cemented

Mud level not at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and observe with moving the drill string and prepare well control related work. Each 10 minutes, mud with density no lower than the drilling weight shall be reversely filled up from the annulus. Each time, the filling up shall be 1~2m^3, or continuous reserve injecting mud with small speed to maintain that the column pressure of mud in wellbore is higher than the formation pressure. POOH to casing shoe or safety interval for observation. Additional treatment shall be made as per the situation of loss of circulation.

Mud level at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and move the drill string to observe with well control well prepared. When the well is static, pull the bit upward to the casing shoe or safe interval for observation. If no anomaly is observed, circulate with slow pump rate. The measures such as plugging back job, POOH or condition mud etc. shall be carried out as per the situation of loss of circulation. If loss of circulation occurs to pay zone, mud weight shall be acquired by static pressure. After pulling out to casing shoe and observing for one tripping cycle, when no more loss of circulation occurs after running to bottom, drilling operation can be resumed.

Well control for E logging, cementing and DST

Before E logging, down hole condition shall be stable and static. If it takes long time for electrical logging, one more condition trip shall be conducted between logging. Shooting nipple is required for logging. Logging service shall be furnished with tools for cutting the cables.

Continuously fill the well during performing wireline logging, and keep closely monitoring for the well via trip tank.

Before running casing, ensure the well is in static condition without any flowing. Pipe ram shall be changed to casing ram with proper size. Casing circulating head or crossover sub between casing and in used DP shall be kept at rig floor at all times during casing running.

The hydrostatic pressure shall be kept balance with formation pressure during whole cementing job to avoid blowout due to loss of circulation, loss of balance for pressure and cement weight loss during WOC.

12.7 HEALTH

This well is committed to develop for oil in a responsible manner that promotes employees' and local inhabitants' health.

A trained Paramedic and proper MEDIVAC equipment and ambulance shall be stationed on the drilling rig during the drilling campaign.

An emergency medical evacuation plan shall be coordinated, and shall keep in touch with emergency response centres manned by qualified doctor 24-hours per day in Ahwaz city.
Medical emergency report and investigation shall be performed as soon as possible in accordance with Medical Emergency Procedures.

12.8 SAFETY

Due to high concentration of H2S in Sarvak, Kazhdumi and Gadvan reservoirs, Make detail H2S operation procedures, in particular, and carry it out strictly.
Lifesaving equipment like SCBA, Escape Mask, H2S detectors (Multi portable and personal detector) and implement the H2S Services system during drilling activity should be provided.
All lifting equipment and cranes must be certified by approved 3rd party (approved by Institute of Standards & Industrial Research of Iran)
All electrical equipment should be designed for such locations, and listed by nationally recognized testing laboratory. All wiring components and electrical equipment should be maintained in accordance with the manufacturers' recommendation.
PSSR (Pre-Start-up Safety Review) before spud-in and penetrating into oil and gas formations are mandatory by contractor under the supervision of Company and approved by Company.

12.9 ENVIRONMENT

Requirement on location

The access road and well location construction shall be accorded with construction program of well location. It minimizes any impacts on the local residents, prevent all soil contamination, prevent all ground water contamination, and prevent all surface water contamination. Cutting and waste water pit shall be lining and drainage system shall be good and to avoid farm and season canal contamination.
Management regulation for transportation truck, drilling operation machine and tools shall be carried out strictly to avoid damage farm and trees, etc.
Well site shall be flat and clean. All the equipment, materials, drilling tools shall be put in order.
Control the rig noise to decrease the influence to surroundings while drilling at the inhabitation area.
Contractor shall perform on-site segregation of wastes prior to transfer to the transfer area and/or delivery to the transportation Contractor in accordance with the following waste hazard classification.
Non-Hazardous Wastes (Re-usable, Recyclable and Recoverable Wastes)
Ferrous and non-ferrous metals, plastics, and wood wastes, PETs and potentially paper and glass.
Materials shall be separately segregated and provision may be made for size reduction and containerization equipment.
The Contractor/Generators are encouraged to seek for recycle/re-use alternatives based on the Local available facilities, for final disposal of these materials.

It should be noted that landfill disposal of metal waste materials and wood wastes shall be prohibited. For wood wastes re-use of and/or salvage of commercial timber, or deliver/release to.
Local community are recommended.

Hazardous Wastes

Hazardous Waste, such as clinic waste, oil waste, waste produced by painting, insulation works shall be duly segregated.

Facilities of environment protection supplied to well site

Mud purifying circulation system.
Cooling water spraying and circulating system for drilling pump.
Special recovery tank for deposed oil.
Reserve house for valuable mud chemicals.
Automatic mud weighting supply system.
BOP stack is equipped according to drilling design.
Define a separation area for gathering all waste according to waste management procedure

Requirement of environment protection on location

Sewage pit Area of the sewage pit: 1500 m2.

Prior to spud, mud pit and sewage pit must comply with the standard. Install cooling water spraying and circulating system for drilling pump and sewage circulating system for cleaning rig floor and equipment.
Install BOP stack and pressure test shall be operated according to the drilling design to avoid blowout.
Install oil supply lines and check whether there is any leakage.

Management of environment protection on location

In drilling operation keep mud and sewage from overflowing. If it happens, clean up timely.
Mud chemicals shall be stored according to the standard and shall not be scattered around well site. If mud chemicals scatter while loading or unloading, clean it up at once. The sack of mud chemicals shall be recovered. Do not dispose it on site.
Enhance water management at well location and do not wash equipment with fresh water. Stop water flowing for long time. Overhead tanks shall be utilized.
Oil management shall be carefully at the well site. No oil dirt on site, no dripping and leaking at oil tank and manifold system area.
After the well is completed, reinforce the sewage pit solidify the disposed mud and have environment procedure fulfilled.
Install fuel supply lines and keep them in good condition.
All the drilling fluid materials shall be recovered and shall not be left at the well site after completion.
The environmental protection personnel from contractor and operator side shall check and accept the well site environment according to the regulation after completion. Only well site environment is qualified, the well can be delivered.

Transition Area for Hazardous Waste shall also be provided by each Contractor in his allocated area, Company shall approve each area before use as well. The transition area of hazardous waste shall be:
Paved, fenced, marked and illuminated
Equipped with appropriate Fire Extinguisher and other emergency response equipment (e.g. Eye-Wash and Emergency Shower, appropriate First Aid kits, neutralizer substances etc.).
In compliance with the requirements of handling, transportation and storage of hazardous waste (based on usage chemical MSDS).
Configured to accommodate a safe physical separation of incompatible waste materials (e.g. acids, and alkalis, organic materials and strong oxidizing agents).
Capable enough to provide storage for the entire range of hazardous wastes generated during the work.
The area shall be provided with Spill Containment and means of preventing run-off to and from the area.
All waste containers to be stored shall be appropriately labelled.
Hazardous Waste shall be transferred in a timely manner, within a maximum period of 48 hours, for any period in excess of this permission shall be granted from Company and/or PEDEC.

12. H_2S Monitoring & Prevention Measure

13.1 Monitoring H2S

Somewhere on the well site is apt to accumulate H2S, such as mud pit, shale shaker, and drilling platform, should be installed one set of. Make sure H2S monitor and alarm system can be used. Alarm threshold setting for H2S warning indicator: the first warning level is 10mg/m3, unable warning sound, just show present value of H2S content to operators. The second warning level is 20mg/m3, when the H2S content in air exceed 20mg/m3, H2S warning indicator can give an alarm automatically, and warning sound should be heard by everyone work in well site. The audible alarm should be installed on Kelly board.

Equip enough gas masks on drilling platform, shaker screen (total amount should large than 24), and equip 12 sets of 30min positive pressure Air Aspirators 2 sets of portable H2S monitor and appropriative H2S warning indicator. Arrange a person specially assigned for taking charge all the protective apparatus. This man must do periodical check to keep these equipment in a good condition.

H2S monitor should be done weekly check and compulsory examination.

After drilling into the pay zone, strengthen the detection of H2S content in mud.

13.2 Measures for H2S prevention

Windsocks shall be set on the drill floor and in the monsoon direction or/and prevailing wind direction of Azadegan oil field, off well site. In emergency, employees must follow the evacuation procedure.

Explosion-proof blowers shall be installed around the rig, shale shaker, and drilling fluid tanks to blow away toxic and flammable gas at the well site.

The rig crew in the H2S area shall be furnished with H2S monitors and adequate protective apparatus. Everybody shall be able to use, maintain, and check the apparatus.

The technicians of rig crew are responsible for H2S proof training and the tool pusher is in charge of the supervision and examination.

The pH value of drilling fluid in the H2S area shall be kept over 9.5. Pay attention to measurement of H2S concentration in drilling fluid. The H2S concentration shall be kept at less than 50mg/cm3 by using sulphur removal agents and degassers.

Personnel safety and protection shall be carried out according to SY/T5087 during operations in polluted areas where H2S content exceeds the critical concentration.

When the rig crew makes decision to bleed and fire because well control cannot be performed due to well site conditions, operations shall be performed according to procedures in SY/T5087.

Once blowout occurs, the incident shall be reported to the higher level department as soon as possible and firefighting trucks, ambulances, medical care personnel and technicians shall be on duty at the well site.

After the blowout is controlled, H2S concentration should be measured in the places where H2S is possible to gather.

The personnel are not allowed to access until the H2S concentration is below the critical concentration.

13. Handover Data

The handover document should contain the following well construction information:

- Pre-spud Inspection Report
- Daily Drilling Report
- Daily Mud Report
- Daily Mud Logging Report
- Completion Tally
- Completion running Report
- TMS data (Completion)
- Dry Test report
- Test Charts, X-Mass tree Installation & test Report.

14. Requirements

Item	Description	QTY.	Unit	Supplier
1	TBG hanger Running tools	1	Ea.	PEDEC-PPA
2	BPV with inserting & retrieving tools	1	Ea.	PEDEC-PPA
3	TWCV with inserting & retrieving tools	1	Ea.	PEDEC-PPA
4	Plug for No-Go Landing Nipple 2.635" Complete with Running & Pulling Tools.	1	Ea.	PEDEC-PPA
5	Prong for No-Go Landing Nipple 2.635" Complete with Running & Pulling Tools.	1	Ea.	PEDEC-PPA
6	Protection Sleeve for Tubing Retrievable Safety Valve Complete with Running & Pulling Tools	1	Ea.	PEDEC-PPA
7	Lubricator for BPV	1	Ea.	PEDEC-PPA
8	Shifting Tools for SSD	1	Ea.	PEDEC-PPA
9	Plug for SSD 2.75" complete with running & pulling Tools	1	Ea.	PEDEC-PPA
10	Prong for SSD 2.75" with inserting & retrieving tools	1	Ea.	PEDEC-PPA
11	SSSV Secondary Valve with lock mandrel (Rig less Job)	1	Ea.	PEDEC-PPA
12	Exercise Tools (Rig less Job)	1	Ea.	PEDEC-PPA
13	Punching Tools (Rig less Job)	1	Ea.	PEDEC-PPA
14	Lock Open Tools (Rig less Job)	1	Ea.	PEDEC-PPA

15	Running & pulling Tools for SSSV Secondary Valve with lock mandrel (Rig less Job)	1	Ea.	PEDEC-PPA
16	Running & pulling tools for Wire Line Retrievable Safety valve (Rig less Job)	1	Ea.	PEDEC-PPA
17	8-1/2" Rock Bit	1	Ea.	MSA
18	6" or 5-7/8" Rock Bit	2	Ea.	MSA
19	6" or 5-7/8" Mill	4	Ea.	MSA
20	3-1/2" Mill	4	Ea.	MSA
21	Taper Tap Suitable for MODEL PREMIER PRODUCTION PACKER SIZE: 7"OD:5.827” MIN ID:2.880”	2	Ea.	MSA
22	Spear Suitable for MODEL PREMIER PRODUCTION PACKER SIZE: 7"OD:5.827” MIN ID:2.880”	2	Ea.	MSA
23	Kazhdumi Completion String 7.5K	1	set	PEDEC-PPA
24	7" Scraper	1	Ea.	OEOC
25	9-5/8" Scraper	1	Ea.	OEOC
26	5" DP,HWDP,8-1/2" DC,	as required		OEOC
27	Well Head Services	as required		Service Co.
28	Surface Well Test Services	as required		Service Co.
29	Completion Services	as required		Service Co.

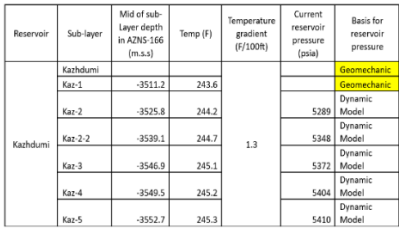
30	Slick line Services	as required		Service Co.
31	Casing Running Services	as required		Service Co.
32	Wireline Logging	as required		Service Co.
33	Dry Test, DST, RTTS	as required		Service Co.
34	Cementing Services	as required		Service Co.
35	Stimulation (Coiled Tubing, Acidizing and N2) Services	as required		Service Co.
36	H2S Services	as required		Service Co.
37	Mud Logging Services	as required		Service Co.
38	Rig Services	as required		Service Co.
39	Drilling & Completion Fluid Services	as required		Service Co.
	Note: Quantity & Spec to be conformed with contract's SOW. Note: In case of any shortage or discrepancy, the contract's SOW will be reference.			

Table 12: List of Requirements

15. Revision Corrections

Drilling Engineering							
1	note ، قسمت Contingency #06در جدول شکست زمانی، مربوط به تهیه ابزار توسط کارفرما حذف گردد.	28-31	CM	حذف گردید			AP
۲	با توجه به اینکه در جدول شکست زمانی Contingency 08 آورده نشده و گرفتن کروژن لاگ جزو برنامه میباشد حذف گردد.	۳۱	CM	در پیوست ۷ قرارداد نهایی شده است.			AP
3	و CBS میبایست طبق contingencyمدت زمان عملیات + صفحه ۷ یکسان سازی گردد. به دلیل اینکه پیمانکار از ویرایش ۰۱ این مدت را از ۴۴ روز به ۷۷ روز تغییر داده است لذا در صورت عدم باشد. ارائه صورت جلسه، این تعداد روز مورد تائید مشاور نمی	28	CM	در پیوست ۷ قرارداد نهایی شده است. مدت زمان تأیید شده عملیات ۴۴,۲ روز میباشد و همچنین ۹۵ در نظر گرفته شده contingencyروز به عنوان است.			AP
٤	ذکر شده است BMW ، BMVدر جدول و متن به اشتباه به جای اصلاح شود.	26-29	CM	تصحیح گردید			AP
۳	با شرح کار متفاوت است. به شرح زیر اصلاح EP و TDمختصات گردد: E.P Coordinates: E:196270.9 N:3472117.8 -3355.2 m TVDss TD Coordinates: E:195961.5 N:3471902.6 -3561.0 m TVDss	۷	CM	• تصحیح گردید			AP
٤	در جدول عمق آستری ۱/۲ ۴" به ۴۱۳۲,۸-۴۲۷۹,۶ اصلاح گردد.	۲۰	CM	• تصحیح گردید			AP

5	آیتم ۶ و ۷ ، در جدول شکست زمانی در صورت موفقیت ادامه فرایند می بایست به آیتم ۲۹ ارجاع گردد و با صفحه ۲۸ یکسان سازی شود.	۲۶	CM	• تصحیح گردید			AP
6	در جدول شکست زمانی، عمق نهایی از ۴۲۷۹,۶ به ۴۲۸۰ و همچنین عمق از ۳۷۶۳ به ۴۲۸۰ متری اصلاح گردد.	۲۶	CM	• تصحیح گردید			AP
7	در جدول شکست زمانی بعد از آیتم اول تحت عنوان note: مورد زیر اضافه شود: If wellhead pressure is not Zero after bleed off, Contractor shall inform Client. Remedial plan to be prepared in advance by Operators and will be followed after client's confirmation.	۲۸-۲۶	CM	• اضافه گردید. • مطابق با پیوست ۷ قرارداد (جدول شکت زمانی تأیید شده)،note مورد نظر وجود ندارد. • اضافه گردید.	در ویرایش ۰۳ مجددا حذف شده است به دلیل اینکه در صفحه ۳۱ .اضافه شود. آورده شده و چرا که می بایست در جدول شکست زمانی نیز آورده شود. جمله مذکور به جدول اضافه گردد.		CM
8	پیشنهاد میگردد در BHA، jar. نیز استفاده گردد.	۳۱	CM	تصحیح گردید			AP
9	برنامه راندن رشته تکمیلی با شرح جزییات و مشخصات کامل ابزار می بایست حداقل ۱۰ روز قبل از عملیات ارائه گردد.		CM	برنامه رانش رشته تکمیلی در زمان مقرر اخذ تایید کارفرما ارسال خواهد شد. جهت			AP

10	فشارهای مخزن به شرح زیر اصلاح گردند: Kazhdumi 5410 psi Kaz 1: 5756 psi Kaz 2: 5472 psi Kaz 3: 5481 psi Kaz 4: 5493 psi Kaz 5: 5495 psi	۱۴	CM	• تصحیح گردید	پیرو بروزرسانی شبیه ساز مخزنی توسط تیم مخزن، فشار های مخزن به شرح زیر می‌باشد.  Reservoir / Sub-layer / Mid of sub-Layer depth in AZNS-166 (m.s.s) / Temp (F) / Temperature gradient (F/100ft) / Current reservoir pressure (psia) / Basis for reservoir pressure Kazhdumi / Kazhdumi / / / / / Geomechanic Kaz-1 / -3511.2 / 243.6 / 1.3 / / Geomechanic Kaz-2 / -3525.8 / 244.2 / 1.3 / 5289 / Dynamic Model Kaz-2-2 / -3539.1 / 244.7 / 1.3 / 5348 / Dynamic Model Kaz-3 / -3546.9 / 245.1 / 1.3 / 5372 / Dynamic Model Kaz-4 / -3549.5 / 245.2 / 1.3 / 5404 / Dynamic Model Kaz-5 / -3552.7 / 245.3 / 1.3 / 5410 / Dynamic Model	14	AP
11	تاریخچه چاه بطور کامل در برنامه آورده شود (مشکلات بوجود آمده 3SV(	۱۷	CM	• اضافه گردید. • اصلاح گردید. • اصلاح گردید.	اصلاح شود.NO-GOعمق NOGO:2769.9 TOF احتما منظور THFمنظور از کلمه میباشد. اصلاح شود.	۱۷-18	CM

۱۲	هنگام برداشتن آسیاب کننده مورد شاخه به شاخه TBDدر برداشتن لوله ها حذف گردد (RIH with 6 or 5-7/8" Mill to 2754m (P/U required DP jt by jt), چون یک مرتبه بعد از برپایی دکل استند شده است. Held(PJSM. Spud the Rig. Stand Back required 5" & 3-1/2" DPs, 5" HWDPs and 6-1/2" DC.(متعاقبا زمان های درنظر گرفته شده نیز تغییر یابد.	۲۶ & ۲۷	CM	• در پیوست ۷ قرارداد نهایی شده است. • به دلیل اینکه بعد از برپایی دکل زمان استند کردن دیده نشده و همچنین با توجه به وجود رشته تکمیلی در چاه، امکان استند کردن لوله‌ها وجود ندارد و متعاقباً زمان استندسازی در هنگام پیمایش با mill در نظر گرفته شده است. • با توجه به اصلاحات صورت گرفته، حذف گردید.	بدون در نظر گرفتن زمان، به دلیل اینکه یک مرتبه بعد از برپایی دکل استند شده (P/U required DP jt by jt),استو جمله در هنگام برداشتن آسیاب کننده حذف گردد. با توجه به توضیحات پیمانکار، که با وجود رشته تکمیلی در چاه امکان استند سازی Stand Backنیست پس جمله . required 5" & 3-1/2" DPs, 5" HWDPs and 6-1/2" DC در زمان . برپایی دکل حذف شود.		CM
۱۳	بعد از پایین رفتن تا روی پستانک نارونده جمله TBDدر retrieve Plug (چون چاه در خط تولید می باشد) حذف گردد متعاقبا زمان های درنظر گرفته شده نیز تغییر یابد.	۲۶	CM	• در پیوست ۷ قرارداد نهایی شده است.			AP
۱٤	به ۱/۱۶ ۴" اصلاح گردد. و با صفحه ۴۷ یکسان XMT capسایز سازی گردد.	۲۷-۳۰	CM	• اصلاح شد.			AP
15	عمق نهایی رشته تکمیلی در تاریخچه صفحه ۴۳ به ۲۷۸۳,۷ اصلاح گردد.	۴۳	CM	• اصلاح شد.			AP

۱۶	پیرو بررسی قرارداد و نیز شرح کار ارسالی پیمانکار، مقتضی است در برنامه Dو Cبراندازی و برپایی تجهیزات تاج سرچاهی سکشن مربوطه قید گردد، لازم به ذکر می باشد مطابق با قرارداد پیمانکار مکلف به تامین تجهیزات مذکور می باشد.		CM	• در شرح کار اعلامی از سوی کارفرما، تعویض سکشن‌های C و D قید نگردیده و همچنین پیمانکار مطابق با جدول شکست هزینه چاه ۱۶۶، کالای مورد تعهد خود را تحویل کارفرما می‌نماید.	منوط به تائید کارفرما.		AN
۱۷	آیتم ۱۷-۱۸ به دلیل قابل بازیافت بودن پکر، آیتم آسیاب کردن contingencyتوپک ثابت، و تلاش برای بازیافت آن و همچنین حذف گردد. #7	۳۱-۲۷	CM	• مطابق با پیوست ۷ قرارداد نهایی شده و امکان تغییر جدول شکست زمانی وجود ندارد. • اصلاح گردید.	فارغ از بحث زمانی، به دلیل قابل بازیافت بودن پکر، جمله آسیاب کردن توپک ثابت، و تلاش برای بازیافت آن و همچنین در آسیاب کردن پکر contingency #7 حذف گردد. و متن اصلاح شود.		CM
۱۸	ذکر شده foloow ، follow در جدول و متن به اشتباه به جای است اصلاح شود	۲۸	CM	• اصلاح شد.			AP
Fluid							
۱	می بایست کمتر از TSSدر سیال تکمیلی مقدار 30mg/L.باشد		CM	ولی در صفحه ۳۷ برنامه این مورد عنوان شده است CM از AP ، بجای statusبا این وجود در قسمت استفاده شده است.			AP